Planning backwards from what students need to learn

Use this when a lesson contains worthwhile activities but the knowledge students should leave with is unclear. Start with the intended learning, then select the explanation, practice and checks that will build it.

How to do it well

1 Specify the endpoint. Describe what students will know or do independently, including the quality of explanation or performance required.

2 Unpack the demand. List the prerequisite knowledge and genuinely new components; identify which missing component would prevent success.

3 Plan the route. Sequence a clear model, supported rehearsal, accurate independent practice and a purposeful variation.

4 Plan decision points. Select questions that will reveal whether students are ready for each increase in demand, with a likely response to common errors.

5 Prepare the resources last. Keep only material that supports the route, and schedule later opportunities to retrieve and apply the learning.

What this sounds like

“By the end of this sequence, you will explain which material is suitable and justify your choice using its properties.”

Classroom example. In History, the endpoint is an explanation of why one factor contributed to an event. Planning begins with the expected causal reasoning and evidence. The teacher identifies the factual knowledge students need, models a causal paragraph, checks the link between claim and evidence, then sets a different factor for practice. The worksheet follows that sequence rather than determining it.

Top tips that make the technique work

Write the final response first. An exemplar exposes the vocabulary, reasoning and component knowledge students will need.

Plan the learning episode. A new idea may need part of a lesson or several lessons; the timetable is not the learning sequence.

Plan the check alongside the task. Decide what a successful response would demonstrate before choosing slides or worksheets.

Keep useful difficulty. Remove logistical obstacles without removing the thinking needed to reach the endpoint.

Plan beyond first success. Build in later retrieval and a less familiar application so today’s performance is revisited.

Do and avoid

Do	Avoid
Align the question, explanation and practice with the same learning.	Planning an activity sequence before defining the learning.
Change the route when a prerequisite is insecure.	Treating a completed product as evidence that every component is understood.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

Barton - Planning a Learning Episode | Barton Lesson planning

Prerequisite knowledge checks

Identifying the foundations for new learning

Use this before new teaching when students need specific existing knowledge to access the explanation. Check the actual prerequisite rather than asking whether students remember last lesson.

How to do it well

1 Identify the prerequisite. Name the small piece of knowledge or process that students must retrieve before the new content can make sense.

2 Elicit it from everyone. Use a short atomised question, mini-whiteboards, call and response or a brief written response. Check competence, not confidence.

3 State the connection. Explain how the new learning builds on what students already know; highlight similarities and differences.

4 Start broad, then narrow. Give the general concept or organising idea before introducing detail.

5 Make students process the link. Use self-explanation, Turn and Talk or a comparison prompt so pupils articulate the connection themselves.

6 Check meaning and revisit. Probe how ideas connect, address weak links, and return to the concept later so the connection remains accessible.

What this sounds like

“Before we explain ionic formulae, show me the charge on a magnesium ion and a chloride ion. Keep your answers private until the signal.”

Classroom example. Before teaching ionic formulae, a Chemistry teacher checks ion charges on boards. Several pupils remember the symbols but confuse charges. The teacher repairs that small prerequisite and checks another pair of ions before modelling charge balance. The new explanation now starts from knowledge the class has shown it can use.

Top tips that make the technique work

Atomise the check. Test a small prerequisite directly, rather than hiding it inside a complex question that makes failure hard to interpret.

Separate recall from application. A remembered definition does not automatically establish that students can use it.

Choose a consequential prerequisite. Check the knowledge that would block today’s learning, not everything previously taught.

Fix the gap before adding complexity. A brief model and a fresh check may be more useful than continuing over an insecure foundation.

Keep the BSCS boundary clear. Spaced retrieval belongs in Do Now; the immediate prerequisite check sits at the opening of Explain.

Do and avoid

Do	Avoid
Gather independent responses before introducing the new demand.	Inferring readiness from confident volunteers.
Make the connection to new learning explicit.	Repeating an entire previous lesson when one component needs repair.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Activating Prior Knowledge | Barton Atomisation

Anticipating misconceptions and planning how to uncover them

Use this when students repeatedly make the same plausible error. Anticipate the reasoning behind that error so the lesson exposes and repairs it rather than merely announcing the correct answer.

How to do it well

1 Start from the intended idea. Write the accurate rule, relationship or distinction students need to understand.

2 Predict the wrong model. Describe what a student might reasonably believe and why that would produce the error.

3 Design the diagnostic prompt. Use a choice, comparison, explanation or short task that distinguishes the correct and incorrect reasoning.

4 Prepare the repair. Select an example and non-example, model or demonstration that addresses the precise difference.

5 Plan the fresh check. Change the context enough to require thinking while retaining the concept you are checking.

What this sounds like

“Someone says heavier objects always fall faster. Which part of that claim needs testing? Explain what evidence would settle it.”

Classroom example. In Geography, a teacher anticipates the belief that weather and climate mean the same thing. Students classify statements about a single day and long-term patterns. Their explanations reveal whether the distinction is secure. The teacher contrasts the timescales explicitly, then gives new statements so pupils apply the repaired distinction.

Top tips that make the technique work

Use real errors. Student work is a better source of plausible misconceptions than invented distractors with no likely rationale.

Separate slips from beliefs. Ask how the answer was reached before assuming a conceptual misconception.

Plan a near-miss. A convincing wrong example can make the critical distinction visible.

Keep the correction focused. Repair the faulty rule or decision rather than explaining every part of the topic again.

Recheck with changed surface features. Students need to apply the repaired idea, not repeat the exact answer they just heard.

Do and avoid

Do	Avoid
Name the misconception each planned answer would reveal.	Calling every incorrect answer a misconception.
Prepare a contrasting example and follow-up question.	Revealing the solution before finding out students’ reasoning.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Diagnostic questions

Diagnostic questions

Planning questions that reveal understanding

Use diagnostic questions when you need to see which misconception or decision is causing difficulty. A well-designed wrong option tells you something useful about students' thinking.

How to do it well

1 Specify the knowledge. Target one important idea, decision or misconception; minimise unrelated reading and calculation demands.

2 Build the options. Write the correct response and plausible distractors, recording the reasoning each distractor would indicate.

3 Protect independent commitment. Explain the response method first, allow silent thinking and collect everyone's choice before revealing any answer.

4 Probe the pattern. Ask selected students for their reasoning, including some who chose correctly; compare alternatives without shaming individuals.

5 Act on the diagnosis. Reteach the identified distinction, use discussion where students have something useful to compare, and ask a fresh related question.

What this sounds like

"Work out your answer first. Now compare it with the four options. Choose privately and prepare to explain which feature rules out the others."

Classroom example. A Maths teacher designs options from likely errors in calculating a percentage reduction: subtracting the percentage as a number, retaining the original amount, or calculating only the reduction. The distribution reveals which decision needs teaching. Students explain a wrong option before completing a fresh calculation, rather than simply being told which letter is correct.

Top tips that make the technique work

Make distractors diagnostic. Each wrong option should arise from a recognisable error you could respond to.

Avoid accidental clues. Option length, grammar or an implausible answer should not give away the correct choice.

Generate before recognising. Where appropriate, hide the options first so students attempt an answer before selecting.

Investigate wrong options. Ask why a student might choose them, or how to alter an option to make it correct.

Inspect correct reasoning too. A correct choice can come from guessing or a faulty method; probe selected explanations.

Do and avoid

Do	Avoid
Use one unambiguously correct answer.	Writing arbitrary wrong options to fill the spaces.
Interpret the pattern of distractors before deciding the response.	Treating the percentage correct as the whole diagnosis.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Diagnostic questions | DistillED - Hinge Questions

Scaffolding

Preparing support and deciding when to remove it

Use a scaffold when a specific barrier prevents students from attempting the intended learning successfully. Keep the goal clear and plan how students will work without the support.

How to do it well

1 Diagnose the barrier. Identify whether pupils lack knowledge, cannot coordinate the steps, are overloaded by presentation, or need a cue to select a known process.

2 Break the process into meaningful steps. Sequence the task so pupils can focus on one decision at a time while still seeing how the parts contribute to the whole.

3 Select a targeted support. Use a model, worked example, prompt, frame, word bank, visual cue, checklist or partially completed response that addresses the diagnosed barrier.

4 Keep the thinking with the pupil. The scaffold should cue attention or structure action; it should not complete the crucial reasoning on the pupil's behalf.

5 Check and adjust. Use responses during practice to decide whether to re-model, retain, reduce or remove the support.

6 Plan the fade before use. Decide which element will disappear first and what evidence will indicate readiness for the next level of independence.

What this sounds like

“Use this opening frame to state your claim. You will choose the evidence and explain the link yourself; the frame does not supply those decisions.”

Classroom example. In English, pupils need to explain how evidence supports a claim. A frame supplies the paragraph structure while pupils choose and interpret the quotation. Once they can build the link, the opening phrases are removed. The scaffold supports expression without replacing the interpretation that the task is intended to develop.

Top tips that make the technique work

Name the barrier. Decide whether the student needs help with knowledge, language, organisation or a particular process step.

Scaffold the obstacle. A vocabulary bank helps with expression; it will not repair an untaught concept.

Make support optional where appropriate. Students who can already work independently should not have to complete unnecessary scaffolding.

Preserve the decision. Give a starting cue or structure without supplying the thinking the task is meant to develop.

Plan the withdrawal. Decide what successful performance would justify removing a cue, worked step or sentence frame.

Do and avoid

Do	Avoid
Match temporary support to the diagnosed need.	Assigning a permanent scaffold by pupil label.
Keep the curricular goal ambitious.	Removing every support at once because time has passed.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Provide Scaffolds for Difficult Tasks | DistillED - Fading Support

Entry routines

Establishing a purposeful start

Use the agreed BSCS entry routine to make the move from corridor to independent learning warm, predictable and purposeful. Students should know where to go, what to prepare and how to begin.

How to do it well

1 Prepare the start. Display the Do Now and have the required resources ready before students arrive.

2 Welcome from the threshold. Establish the agreed queue and entry expectations while monitoring the corridor and classroom.

3 Release to the task. Students go to allocated seats, prepare equipment and begin silently; start the timer from first entry.

4 Maintain the entry position. Greet students and monitor the developing routine until a critical mass is established, using brief precise corrections.

5 Secure independent thinking. Narrate successful actions, complete the register when settled and use the agreed signal before reviewing the Do Now.

What this sounds like

“Good morning. Equipment ready. Straight to your seat and begin the Do Now silently. You have four minutes.”

Classroom example. For a class arriving after lunch, the teacher displays four retrieval questions, has equipment ready and greets students at the threshold. The timer begins as the first pupils enter. Brief corrections address the exact entry action while the majority begin silently. The teacher moves into review only after using the agreed attention signal.

Top tips that make the technique work

Prepare before entry. Display a Do Now that students can start without an explanation and have resources ready.

Keep the threshold position. Welcome students while monitoring the corridor and the developing classroom routine until the class is established.

Start the timer from first entry. Use the agreed three-to-five-minute window and build productive urgency.

Narrate the useful action. Recognise students beginning the task, rather than making general comments about being good.

Rehearse the exact sticking point. If equipment or movement causes repeated delay, practise that component rather than changing the whole routine.

Do and avoid

Do	Avoid
Combine a warm greeting with clear expectations.	Using entry as an unplanned settling period.
Correct briefly and precisely while protecting the start.	Beginning an explanation while students are still moving or gathering equipment.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Solid Entry Routines | Barton - Embedding habits and routines

Teaching and rehearsing classroom routines

Use this when a repeated classroom process is unreliable. Routines become useful when students understand, practise and rehearse the exact actions, rather than hear a rule once.

How to do it well

1 Specify the actions. Break the routine into a short sequence with a clear starting cue and ending condition.

2 Explain and model. State why it matters, demonstrate success and show the common mistake students must avoid.

3 Rehearse together. Practise the process with low-demand content, watching the whole class for the precise point of breakdown.

4 Give focused feedback. Name what worked, correct one component and repeat it until the class can perform the sequence fluently.

5 Maintain the habit. Use the same cue, recognise successful execution and re-teach briefly when the routine drifts.

What this sounds like

“We are practising the move from partner talk to listening. At the signal, finish the word you are saying, stop talking and put your pen down.”

Classroom example. A Music teacher teaches how to move from listening to using instruments. Students practise collecting one instrument, waiting with hands still and starting on the cue. When premature playing reappears, the teacher rehearses that part briefly with an easy rhythm. The musical content can then proceed without competing with an uncertain equipment routine.

Top tips that make the technique work

Give the routine a learning purpose. Explain how it protects attention, safety or time, using language students understand.

Teach the cue and response together. Students need to recognise when the routine starts as well as what to do.

Show a convincing non-example. Make the difference visible without turning a pupil’s mistake into public entertainment.

Practise before content matters. Rehearse with an easy task so students can attend to the mechanics.

Revisit after drift. A short reset after holidays or repeated breakdown is usually better than increasingly lengthy reminders.

Do and avoid

Do	Avoid
Give feedback on an observable action.	Assuming older students already know the routine.
Repeat the part that was not secure.	Replacing the cue every time the routine is imperfect.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Embedding habits and routines | DistillED - Behaviour Management

Signal for attention

Securing readiness before teaching

Use the agreed BSCS signal before instructions, explanations and transitions. Secure readiness before speaking so the beginning of the message reaches the whole class.

How to do it well

1 Prime the transition. State when students will stop and what readiness will look like, before the attention signal.

2 Position visibly. Stand where all students can see you and you can observe the whole room.

3 Use the agreed BSCS signal. Give the current Focus Ready cue consistently and allow students to respond.

4 Pause and scan. Wait for the class to be ready; use a brief, precise correction for the remaining issue.

5 Give the message and check listening. Deliver a concise instruction or explanation, then ask students to retrieve the key action or point.

What this sounds like

“In ten seconds, stop and get Focus Ready.” [Use the agreed BSCS signal.] “Thank you. First, write the title. What is the first action?”

Classroom example. During partner discussion, the teacher primes the stop, moves where all pupils are visible and uses the agreed signal. The teacher waits and corrects two unfinished conversations before giving the instruction. A listening check establishes that students know the first next action. The transition does not begin with several pupils missing the message.

Top tips that make the technique work

Prime the transition. Tell students when they will stop and what readiness will look like before using the signal.

Choose a visible position. Stand where you can see every area and students can see you.

Be seen looking. Scan deliberately; your attention should communicate that you are checking the response.

Wait before academic talk. The signal loses meaning if the explanation starts while conversations or movement continue.

Follow with a listening check. Readiness gives students access to the message; recalling a key instruction shows whether it was received.

Do and avoid

Do	Avoid
Use the current shared BSCS cue consistently.	Speaking over partial attention.
Correct the smallest remaining issue calmly.	Treating eyes towards you as proof that the explanation was understood.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Signal for Attention

Checks for listening

Instructions, explanations and classmates

Use brief checks to establish whether students heard the instruction, key explanation point or contribution from a classmate. This helps distinguish missed information from difficulty understanding it.

How to do it well

1 Plan the message. Identify the exact instruction, definition or relationship students need to retain and make it concise.

2 Establish the routine. Explain why listening will be checked, secure attention and deliver the message without competing activity.

3 Ask for recall. Use question–pause–name, choral response or a brief written response to retrieve what was actually said.

4 Repair the missed point. Re-deliver or reframe it clearly, then check again; ask students to paraphrase a peer's contribution where appropriate.

5 Move to understanding. Give a fresh question requiring application or explanation. Students can repeat accurately while remaining unsure how to use the idea.

What this sounds like

"A catalyst speeds up a reaction without being used up. A catalyst speeds up…? Something that speeds up a reaction without being used up is…?"

Classroom example. An English teacher explains that the claim must be supported with a quotation. A pupil states where the evidence should appear, then another restates the key point from a classmate's answer. Those checks establish receipt. A new paragraph task then tests whether students can select suitable evidence and explain the connection independently.

Top tips that make the technique work

Teach the distinction. Explain that this question checks what was just said; a later task will check whether students can apply it.

Use Explain–Frame–Reframe. State a key relationship, ask students to complete it, then reverse the prompt where the reversal remains accurate.

Check peers' contributions. Ask another student to restate the idea before inviting agreement, challenge or elaboration.

Make the message accessible first. Secure quiet, speak audibly and avoid distributing resources or introducing competing text while delivering it.

Keep corrections supportive. Restate the short message and give another opportunity; hearing, language or processing difficulties must not be treated as deliberate inattention.

Do and avoid

Do	Avoid
Check instructions, explanations and classmates.	Using "Are you listening?" as the check.
Ask a separate application question afterwards.	Assuming successful repetition establishes understanding.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Checking for listening | Barton - Embedding habits and routines

DistillED Habits of attention

Behaviour scripts

Correcting calmly and precisely

Use calm, precise behaviour language to restore learning while preserving relationships. Apply the school behaviour system consistently and keep the response focused on the action that needs to change.

How to do it well

1 Describe and direct. State the observable behaviour and the expected action instead of asking a confrontational 'why' question.

2 Use partial agreement. Acknowledge the pupil's comment without abandoning the expectation: 'Maybe so—and the instruction still stands.'

3 Allow take-up time. Give the direction once, move attention back to teaching and return later to check compliance rather than creating a public stand-off.

4 Focus on primary behaviour. Address what is interrupting learning and, unless serious, avoid escalating a sigh, mutter or eye-roll that follows the correction.

5 Offer directed choices. Give two acceptable, behaviour-based options, state the consequence calmly and follow through consistently.

What this sounds like

"You are talking while I am explaining. Stop the conversation and listen to this step. Thank you."

Classroom example. During silent practice, one pupil starts a conversation. The teacher gives a brief description and direction, allows take-up time and resumes monitoring. A quiet return checks that the pupil has begun the task. When another pupil starts promptly, the teacher recognises the specific action. The correction does not become a public debate.

Top tips that make the technique work

Describe and direct. Name the observable behaviour and the required action, avoiding questions that invite an argument.

Allow take-up time. Give a clear direction, move attention away where appropriate, and return to check compliance.

Use partial agreement carefully. Acknowledge a point without abandoning the expectation: "Even if you were helping, this is silent work."

Focus on the primary behaviour. Avoid escalating a minor sigh or facial expression while correcting the disruption, unless the secondary behaviour is serious.

Make praise precise. Positive narration and recognition should identify the useful learning action rather than publicly rank pupils.

Do and avoid

Do	Avoid
Use brief, predictable language and follow through.	Debating publicly or reacting emotionally.
Begin with the least intrusive effective response.	Inventing consequences outside the agreed school system.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Behaviour Management

Exit routines

Ending lessons calmly and efficiently

Use the agreed BSCS exit sequence to finish learning, restore the room and move students calmly into the corridor. Plan sufficient time for practical equipment and resource return.

How to do it well

1 Stop the academic activity. Secure attention while enough time remains to complete the room's exit routine.

2 Explain resource return. Give one clear instruction and check listening before releasing students to act.

3 Pack in stages. Cue the remaining actions, including resetting equipment and standing ready; keep students who have finished waiting calmly.

4 Check readiness. Scan the room, correct unfinished actions and move to the threshold before dismissal.

5 Dismiss deliberately. Release by the agreed row or group sequence, monitor corridor movement and leave the teaching space ready for the next lesson.

What this sounds like

"When I say go, pass the books to the end of your row and then wait. Go. Thank you. Now pack away and stand behind your chair."

Classroom example. At the end of a practical lesson, the teacher finishes the learning before giving the equipment instructions. Pupils return materials on a clear cue, reset the benches and wait ready. After checking safety and room condition, the teacher dismisses by the agreed sequence from the threshold, maintaining a view of the corridor.

Top tips that make the technique work

End the learning deliberately. Do not let premature packing swallow the final explanation or response.

Front-load each action. Tell students what to do before mentioning an object likely to trigger movement.

Use one release cue. Students should know when to act and when to wait.

Check the room before dismissal. Resource return, workspaces, uniform and equipment need a planned check.

Keep sight of the corridor. Dismiss from the threshold in a sequence that allows you to monitor both spaces.

Do and avoid

Do	Avoid
Use the agreed sequence and concise instructions.	Allowing the bell to become the dismissal cue.
Include room-specific safety and equipment steps.	Giving several competing packing instructions at once.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Embedding habits and routines | DistillED - Behaviour Management

Do Now

Preparing students for learning

At BSCS, Do Now provides a short, independent start using spaced retrieval of previously taught, curriculum-critical knowledge. Keep the routine stable so thinking begins without a fresh explanation.

How to do it well

1 Select curriculum-critical knowledge. Use a spaced retrieval plan across recent and older content. Avoid trivia, isolated detail and one-off quiz questions.

2 Pitch for productive success. Aim for roughly three quarters correct overall: question 1 accessible, questions 2–3 secure for most, question 4 more challenging, plus an optional challenge.

3 Use clear, single-answer prompts. Prefer discrete facts, features, vocabulary, short procedures or carefully bounded explanations. Do not stack recall, explanation and application into one overloaded prompt.

4 Make the routine automatic. Display the task in the same place, use a visible timer, keep the response location consistent, and expect quiet independent thinking without notes.

5 Collect representative evidence. Use mini-whiteboards, Book-to-Board, ABCD cards, call and response or planned indicator pupils rather than volunteers alone.

6 Respond and reschedule. Correct misconceptions, model briefly where needed, use a related recheck, and return to fragile knowledge in a later Do Now.

What this sounds like

“Complete the four core questions silently, without notes. If you are stuck, move on and return. The final question is an optional challenge.”

Classroom example. A Languages Do Now retrieves vocabulary from several earlier lessons, with an accessible first item and an optional translation challenge. Students begin independently without notes. The first teacher lap checks the routine and the second samples answers. A shared vocabulary gap gets a brief repair and is scheduled for retrieval again later.

Top tips that make the technique work

Use four core questions and a challenge. An accessible first item helps everyone start; later items increase the demand without blocking all progress.

Make answers quick to inspect. A bounded response reveals more than an overloaded question with several unrelated demands.

Teach a stuck protocol. Students move to another item and return, rather than waiting with no work completed.

Make two purposeful laps. First secure participation and the routine; then inspect knowledge and likely errors.

Place prerequisites correctly. Check knowledge required for today’s new content at the opening of Explain, rather than replacing spaced retrieval with a starter on today’s topic.

Do and avoid

Do	Avoid
Protect independent attempts before feedback.	Retrieving content students have not yet been taught.
Use a timer and proportionate review.	Turning the Do Now into a long worksheet or lengthy reteaching lesson.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - The Do Now | Barton - Do Now

Retrieval practice

Strengthening recall

Use retrieval practice to strengthen access to previously taught knowledge. Students should attempt to generate the answer from memory before they see notes, prompts or the solution.

How to do it well

1 Select important taught knowledge. Identify the facts, relationships or process steps students need to retain, and write prompts that retrieve them directly.

2 Choose the delay. Revisit recent and older learning so recall involves some effort while remaining achievable.

3 Protect generation. Set a silent independent attempt, keeping solutions hidden and controlling when prompts become available.

4 Give corrective feedback. Compare answers with an accurate model, repair misunderstandings and let students attempt a related item.

5 Schedule the return. Bring fragile knowledge back in later lessons, quizzes or homework, varying the cue while keeping the learning focus clear.

What this sounds like

“Try from memory first. Leave your notes closed. Afterwards we will compare with the model and identify which parts need another retrieval attempt.”

Classroom example. After teaching photosynthesis, a Biology teacher later asks pupils to reconstruct the word equation from memory. Students attempt before the model appears. Feedback clarifies a common confusion between a reactant and product. A later question uses a different cue so pupils retrieve the relationship again rather than remember the position of words on a slide.

Top tips that make the technique work

Match retrieval to the knowledge. Use definitions, explanations, diagrams, short procedures and comparisons where each rehearses the intended content.

Teach before testing recall. Repeated failure on untaught or poorly understood material will not repair the original explanation.

Keep the attempt low stakes. Errors reveal what to revisit; avoid creating anxiety through public rankings.

Show useful feedback. Include the reasoning or worked steps where a final answer alone would leave the error unresolved.

Use later retrieval. A correct answer immediately after teaching does not establish durable recall.

Do and avoid

Do	Avoid
Require an independent attempt before revealing answers.	Replacing retrieval with rereading.
Revisit important gaps after feedback.	Counting copied corrections as successful recall.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Retrieval Practice | Barton - Low-Stakes Quiz

Spaced retrieval

Revisiting knowledge over time

Use spaced retrieval when students can answer soon after teaching but struggle weeks later. Plan repeated returns to important knowledge rather than treating a topic as finished.

How to do it well

1 Identify the durable knowledge. Select the curriculum content students will need to use again, especially prerequisites for later topics.

2 Map return points. Include an initial practice opportunity and later retrieval across lessons, quizzes and homework.

3 Vary the cues carefully. Ask for the same knowledge through a definition, example, diagram or application without suddenly testing untaught content.

4 Respond to forgetting. Explain the important error, provide an accurate model and give another independent retrieval opportunity.

5 Adjust the next return. Revisit fragile items sooner and maintain occasional retrieval of secure knowledge; do not use a rigid interval regardless of success.

What this sounds like

“This question returns to a topic from three weeks ago. Try it from memory before using the worked solution.”

Classroom example. An English team maps several returns to important terminology across a unit. Pupils first retrieve definitions, later identify examples and later explain the term in a new text. Incorrect responses receive feedback before another attempt. The returns are spaced across learning rather than concentrated into a single revision lesson before the assessment.

Top tips that make the technique work

Space the learning, not just the timetable. A later prompt should require students to recall the idea again; displaying the answer beforehand removes that opportunity.

Expect effort. Some forgetting makes retrieval demanding, but repeated near-total failure suggests the delay or prompt needs adjusting.

Keep a curriculum record. Track what has been retrieved and which ideas need another return so selection is deliberate.

Combine spacing with feedback. Correct errors before the next attempt; do not repeatedly rehearse the same wrong answer.

Distinguish spacing from interleaving. Spacing concerns time between encounters; interleaving concerns mixing problem types that require selection.

Do and avoid

Do	Avoid
Plan returns across lessons and units.	Leaving all revision until the end of a unit.
Include both recent and older important knowledge.	Assuming one delayed quiz is sufficient for every student.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Successive Relearning | Barton - Four key retrieval opportunities

Low-stakes quizzes

Making retrieval accessible and challenging

Use a recurring low-stakes quiz to revisit knowledge across topics and identify what each student needs to retrieve again. The review matters as much as the score.

How to do it well

1 Plan the content. Select curriculum-critical material across recent and older topics, including short procedures and conceptual distinctions.

2 Establish the routine. Keep timing, response location and expectations predictable; students attempt without copying or consulting solutions.

3 Commit before review. Where useful, ask students to rate confidence in each answer before they discover whether it is correct.

4 Review selectively. Show accurate worked solutions and spend most explanation time on high-value gaps and confidently held errors.

5 Create future practice. Students make Review Cards or equivalent retrieval prompts, then revisit them in later quizzes, Do Nows or homework.

What this sounds like

“Answer independently. Before we review, rate your confidence. We will begin with important errors and turn those into questions you can practise again.”

Classroom example. A weekly Maths quiz includes recently practised work and older content. Pupils answer alone and rate confidence before review. The teacher shows worked solutions and spends most time on an important confidently held error. Pupils turn that item into a Review Card and encounter related questions again in later quizzes and homework.

Top tips that make the technique work

Mix recent and older material. Use a deliberate retrieval plan rather than a quiz on whichever questions are easiest to find.

Record confidence before feedback. Confidently held errors can guide the order of review and Review Card creation.

Reveal complete solutions. Students need to see the important decision or method, not merely the final answer.

Keep scores in perspective. The quiz is a learning opportunity; avoid public comparison or turning every attempt into a high-stakes assessment.

Convert errors into retrieval. A corrected script is harder to self-quiz from than a question on the front of a Review Card.

Do and avoid

Do	Avoid
Protect the independent attempt and feedback sequence.	Only testing this week’s content.
Plan a later return to corrected knowledge.	Giving a score without explaining the relevant errors.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Low-Stakes Quiz | DistillED - Retrieval Practice

Do Now feedback

Checking responses and addressing gaps

Use a proportionate review so Do Now both strengthens recall and informs teaching. Gather the answers first, then decide which items need confirmation, discussion or a brief repair.

How to do it well

1 Collect independent responses. Use boards, a bounded written answer or a deliberate sample before revealing the solution.

2 Identify the pattern. Distinguish isolated slips, missing knowledge and a shared misconception; note students whose response remains unclear.

3 Choose the smallest useful response. Confirm secure items briefly, discuss genuine differences or re-model the point causing widespread difficulty.

4 Recheck the repaired idea. Use a fresh question with comparable demand and enough change to require independent thinking.

5 Reschedule the knowledge. Record the item for later retrieval and follow up individual gaps during subsequent teaching or practice.

What this sounds like

“Most of us have question one secure, so I will confirm it briefly. Question three shows two different ideas; let us compare those before trying a related question.”

Classroom example. A class answers most Do Now items accurately, but several confuse evaporation with boiling. The teacher confirms secure items briefly, contrasts the two ideas and asks a fresh question about a puddle drying. That response informs the next move. The distinction returns in a later Do Now instead of consuming the entire starter today.

Top tips that make the technique work

Do not give every item equal time. A nearly universal correct response rarely needs the same explanation as a common misconception.

Keep the solution hidden while scanning. An early answer changes what later responses can tell you.

Ask for a fresh response after repair. A slightly changed question shows more than asking students to copy the corrected answer.

Use discussion when it has a purpose. Compare different plausible responses, then return to an individual answer so the evidence remains usable.

Record fragile knowledge. Revisit it in future retrieval rather than trying to solve every long-term issue in today’s starter.

Do and avoid

Do	Avoid
Choose the response from the pattern of answers.	Asking only volunteers for their answers.
Keep the review concise and consequential.	Automatically reteaching every question in full.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Do Now | Barton - A model for responsive teaching

Small steps

Explaining new ideas in manageable chunks

Use small steps when several interacting decisions make new content hard to follow. Teach a coherent component, rehearse it and then connect it to the next part.

How to do it well

1 Set the micro-goal. Signal the one small piece of learning students are about to tackle and link it to prior knowledge.

2 Model the step. Demonstrate clearly and think aloud so students see what success looks like and why the process works.

3 Guide immediate practice. Give students a supported attempt straight away using prompts, part-completed examples or peer rehearsal.

4 Check and correct. Gather responses from all students, diagnose patterns of error and reteach before mistakes become established.

5 Build to the next step. Move on when most are secure, connect the mastered component to the next and gradually combine steps into a complete process.

What this sounds like

“For this example, we are only choosing the correct equation. Once that decision is secure, we will substitute the values.”

Classroom example. For a multi-stage calculation, students first choose the relationship, then substitute values, then manage units. The teacher checks each consequential decision before combining the process. Once the separate components are secure, pupils complete a whole calculation with fewer cues. The small steps preserve the overall purpose of the method.

Top tips that make the technique work

Keep the whole idea visible. Students should know how the current step contributes to the complete process.

Make a step teachable. “Complete question one” is not a learning component; “choose the correct relationship” may be.

Check before adding a dependent step. An error in the first decision can make every later answer misleading.

Use immediate rehearsal. Do not save all student practice until the end of a long explanation.

Recombine deliberately. After components are secure, ask students to complete the process without separate cues for every move.

Do and avoid

Do	Avoid
Use coherent micro-goals and consistent terminology.	Fragmenting the content until the overall meaning disappears.
Build back towards independent whole-task performance.	Equating small steps with slow teacher speech.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Small Steps | Barton - Atomisation

The visualiser

Live modelling step by step

Use the visualiser to show how a response is constructed, including decisions, annotations and revisions. Students see the sequence that a finished example can hide.

How to do it well

1 Prepare the surface. Position the visualiser, test the view and lay out the page so key work remains visible without constant movement.

2 State the goal. Explain what students should attend to and connect the model to the intended learning.

3 Construct the response. Write, draw or demonstrate one coherent section, narrating the consequential choices rather than every physical action.

4 Pause deliberately. Check listening, let students read an annotation or ask them to explain a decision before adding complexity.

5 Move into practice. Give a distinct but related example, keep the model available initially and gather everyone's first attempt before releasing longer work.

What this sounds like

"Watch this first decision without copying. I am choosing this evidence because it supports the claim. Now pause and identify the feature that made it useful."

Classroom example. A Product Design teacher sketches a joint under the visualiser, narrating why each feature is positioned. Students watch the unfamiliar construction first, then annotate an available model. Their guided task uses a different set of dimensions. The visualiser makes the drawing sequence and decisions visible without turning the lesson into simultaneous copying.

Top tips that make the technique work

Check the back row view. Focus, contrast, orientation and text size matter before the modelling starts.

Rehearse the model. Decide where the difficult choice, likely error and pause will occur; do not improvise the whole process.

Separate watching from copying. When the decision is novel, protect attention to the model and provide recording time afterwards.

Control where students look. Point to the relevant line or feature, and step away from the display when you want attention on you.

Preserve a useful model. Keep it available during early practice, then remove it when students are ready to attempt independently.

Do and avoid

Do	Avoid
Make decisions and revisions visible.	Projecting an unreadable polished answer.
Follow the model with a different guided example.	Talking continuously while students try to copy every line.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - The Visualiser The Complete Guide | Barton - I Do

DistillED Live drawing

Think aloud

Making expert thinking visible

Use a think aloud when the important learning sits in the choices an expert makes. Show how you plan, monitor and evaluate, including why you reject an attractive alternative.

How to do it well

1 **Clarify the task.** Read the demand and state the goal before starting.

2 **Connect to prior knowledge.** Identify a similar task, concept or strategy and explain the link.

3 **Plan and predict.** State the intended approach and anticipate where difficulty or error may occur.

4 **Verbalise reasoning.** Explain why each decision or method is chosen.

5 **Monitor and adjust.** Pause, notice when something looks wrong, check and troubleshoot visibly.

6 **Reflect and evaluate.** Summarise what worked, what could improve and what to do next time.

What this sounds like

“My first thought is to use this quotation, but it describes the setting. I need evidence about the character’s motive, so I will choose this line instead.”

Classroom example. An English teacher models selecting evidence. The think aloud explains why an attractive quotation does not support this particular claim, then identifies a better one. During guided practice, pupils name the feature that makes their own quotation relevant. The model reveals a decision they can reuse rather than just the chosen sentence.

Top tips that make the technique work

Script the consequential decisions. Identify a few places where your reasoning adds value, rather than narrating every movement.

Model monitoring. Show how you notice an implausible answer or a weak sentence and decide what to check.

Explain a rejected option. Students often need to hear why a familiar approach does not fit this task.

Make the thinking transferable. Name the clue, rule or criterion students can use on another example.

Hand over the self-talk. During guided practice, ask students to articulate the same planning or checking move in their own words.

Do and avoid

Do	Avoid
Narrate planning, checking and revision.	A stream of unrelated inner commentary.
Keep the model focused on the learning decision.	Pretending every expert decision is immediate and certain.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Thinking Aloud | Barton - I Do

Examples and non-examples

Clarifying what counts

Use examples and non-examples to make the defining feature of a concept clear. Carefully chosen near-misses help students stop relying on superficial appearance.

How to do it well

1 Identify the critical feature. Name the single feature that determines whether something belongs to the concept.

2 Choose a near-miss. Use a non-example that shares the example's surface features but removes the critical feature. Avoid random or obviously wrong cases.

3 Alternate and vary. Show an example, then a related non-example; change one feature at a time so the contrast is visible.

4 Include boundary cases. Use cases that are just inside or just outside the concept so students must apply the definition, not guess from appearance.

5 Pause, commit, reveal. Ask what changed and what stayed the same; have every student decide before revealing and explaining the answer.

What this sounds like

"Both diagrams look similar. Which is a circuit, and which feature decides? Explain the smallest change that would make the other one a valid example."

Classroom example. A Maths teacher contrasts a valid rectangle with a near-miss that shares several visual features. Pupils identify the defining properties rather than rely on familiar orientation. A tilted valid example and a plausible invalid case follow. Students justify each classification, making the concept boundary clearer than a page of identical conventional examples.

Top tips that make the technique work

Choose close contrasts. A near-miss should share enough features with the example to make the boundary worth thinking about.

Change one important feature. Too many simultaneous changes make it difficult to see what determines membership.

Include unusual valid cases. Prevent students from treating the first familiar example as the definition.

Ask for the criterion. "It looks right" is less revealing than a reason linked to the critical feature.

Return to the definition. After comparisons, students should be able to describe what counts and test a new case against it.

Do and avoid

Do	Avoid
Use boundary cases and independent decisions.	Obviously irrelevant wrong examples.
Connect the judgement to the defining feature.	Changing several features so the critical contrast disappears.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Non-Examples

Dual coding

Using diagrams and words together

Use words and visuals together when a diagram makes a relationship, sequence or structure easier to understand. Select the representation for its explanatory purpose.

How to do it well

1 Choose the relationship. Decide what the visual will make clearer: position, sequence, cause, comparison or structure.

2 Design the representation. Use simple shapes, consistent conventions and labels placed close to the relevant features.

3 Explain in stages. Point or trace while introducing each important part, keeping spoken language aligned with the visual.

4 Allow processing. Pause for reading, annotation or a short explanation; do not introduce another demand while students inspect new information.

5 Rehearse the meaning. Ask students to interpret or reconstruct the visual and explain the relationship in words before using it in a new task.

What this sounds like

“Follow this arrow as I explain the transfer. Now describe what it represents. Which part of the diagram shows where the energy is stored?”

Classroom example. In Science, a simple circuit diagram appears in stages as the teacher traces the route. Labels are placed beside components and reading time is separated from narration. Pupils then explain what a connection represents and complete a missing part. The diagram carries the relationship; decorative images would add no useful information.

Top tips that make the technique work

Use a meaningful visual. A diagram should explain the content rather than decorate the slide.

Put labels near their referent. Students should not have to search between a distant key and the relevant feature.

Build gradually. Reveal a structure in the same order as the explanation so attention follows the intended sequence.

Avoid competing language. Pause narration while students read substantial new text.

Ask students to use the representation. Labelling, tracing, explaining and recreating reveal more than simply copying the picture.

Do and avoid

Do	Avoid
Align speech, pointing and the displayed feature.	Using decorative pictures as evidence of dual coding.
Remove visuals that add no explanatory value.	Crowding the diagram with every possible label at once.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Dual Coding

Explicit vocabulary teaching

Developing keystone vocabulary

Use explicit vocabulary teaching when disciplinary language is essential to understanding and expressing an idea. Students need to hear, say, use and revisit the word in context.

How to do it well

1 Identify the essential term. Choose vocabulary needed to understand the content or produce a precise response.

2 Say and explain it. Model pronunciation, give a concise definition and connect it to a familiar idea without distorting the meaning.

3 Show boundaries. Use examples, non-examples and any important contrast with the everyday meaning.

4 Rehearse active use. Students say, write and apply the term in a sentence, explanation or comparison; prompt precision where necessary.

5 Return later. Include the term in retrieval and new contexts so students can recognise and use it without the original prompt.

What this sounds like

“In science, work means energy transferred when a force moves an object. Use that definition to explain which of these examples counts as work.”

Classroom example. In Business Studies, pupils learn the disciplinary meaning of profit and contrast it with revenue. They use the terms in a short explanation and classify examples before calculating. Later retrieval asks for the distinction through a new scenario. Correct use depends on meaning rather than recognising a copied definition.

Top tips that make the technique work

Select keystone vocabulary. Prioritise words that unlock the current concept and recur in later learning.

Address familiar meanings. Words such as “work”, “power” and “significant” can have a different disciplinary meaning.

Use morphology when useful. A prefix or root can support meaning, but it should not replace the subject definition.

Require use in a full idea. Correct pronunciation alone does not show that a student understands the term.

Revisit through varied cues. Ask for a definition, an example, a non-example or an explanation using the word.

Do and avoid

Do	Avoid
Teach a precise meaning and examples.	Giving a long list of definitions to copy once.
Build repeated use into explanation and practice.	Accepting the correct word in an otherwise inaccurate explanation.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Tips for Teachers - Alex Quigley on keystone vocabulary

I Do We Do You Do

Moving towards independence

Use I Do, We Do, You Do to move students from observing a clear model to accurate independence. The phases may repeat; readiness determines the handover.

How to do it well

1 I Do the model. Explain one coherent example, revealing the method and the reasons for key choices.

2 We Do a related example. Students supply a decision or step while the teacher guides, checks and corrects.

3 Check the whole class. Use boards, a brief written response or a practical demonstration rather than assuming readiness from volunteers.

4 You Do a small attempt. Give a closely matched task that students complete independently, with success criteria and any necessary temporary support.

5 Increase responsibility. Remove prompts and add variation when performance is sufficiently secure; return to guidance if common errors persist.

What this sounds like

“Watch my example first. You will now try a different example one step at a time. Once those decisions are secure, you will complete the next one independently.”

Classroom example. A teacher models one paragraph, guides a second response using different evidence and collects everyone’s opening claim. Where the claim is secure, pupils attempt another paragraph independently. A common gap in the explanation leads to a short return to modelling. The phases repeat according to readiness rather than a fixed allocation of minutes.

Top tips that make the technique work

Use different examples. We Do should require the same underlying learning without being a copy of the I Do.

Keep the model visible initially. Support comparison without allowing students simply to substitute a superficial detail.

Make We Do genuinely participatory. Every student attempts a step; a few pupils supplying answers is not class-wide rehearsal.

Release a short first task. Inspect early independent work before students repeat the method across many items.

Loop back when needed. Re-model a sticking point rather than persisting with independent work students cannot yet complete.

Do and avoid

Do	Avoid
Use evidence to decide the release.	Treating phases as fixed time slots.
Preserve the connection between model and practice.	Making You Do substantially harder without teaching the new demand.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Explicit Instruction | Barton - We Do

Guided practice

Checking readiness for independent work

Use guided practice when students have seen an explanation but are not yet ready to perform independently. Keep the demand closely aligned with the model and inspect each important decision.

How to do it well

1 Frame the rehearsal. Explain that this is supported practice, not a test. Name the specific feature or step being rehearsed and keep the first burst short.

2 Model and co-work. Show the process briefly, then complete a closely matched example with pupils supplying decisions or steps.

3 Prompt student explanation. Ask pupils to explain why a step is valid, how it connects to the goal, or what would happen if it changed.

4 Use all-in practice. Require every pupil to respond through mini-whiteboards, paired explanation, call and response, a short written step or another visible routine.

5 Give immediate feedback. Name the precise success or misconception, model the correction if needed, and have pupils make the change straight away.

6 Release gradually. Move from teacher model to co-work, prompted attempt, lightly supported attempt and then a small number of independent items.

What this sounds like

“Try only the first decision. Show me which relationship you will use and why. We will check that before moving to the calculation.”

Classroom example. In Languages, the teacher models a sentence and guides a new one with different vocabulary. Everyone writes the first verb choice before the class continues. A mistaken tense is repaired immediately. Students then complete a short independent set, so the handover follows visible success rather than the contribution of a few fluent speakers.

Top tips that make the technique work

Use a separate example. Students should apply the model, rather than reproduce the answer they just watched.

Check one step at a time. A final answer can hide where the reasoning broke down.

Keep everyone active. Brief board responses, written steps or rehearsed explanations make the rehearsal class-wide.

Explain why the step works. Correct performance and understanding of the principle both matter.

Extend guidance when necessary. Independence is the destination, but the release should follow sufficient accuracy rather than elapsed minutes.

Do and avoid

Do	Avoid
Rehearse the target learning with immediate feedback.	Co-working solely with the most confident pupils.
Release responsibility progressively.	Leaving students to practise a common error repeatedly.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Guide Student Practice | Barton - We Do

Completion problems

Bridging modelling and independence

Use completion problems to bridge a full worked example and an independent solution. Keep part of the successful structure visible while students supply the missing decisions.

How to do it well

1 Model the full example. Complete the method from start to finish and explain the purpose of each important decision.

2 Fade the final step. Leave only the ending incomplete and ask pupils to finish it while the earlier structure remains visible.

3 Prompt the principle. Ask pupils to explain the rule, relationship or criterion that makes their completion valid.

4 Increase the blanks. Remove more of the example so pupils complete progressively larger and earlier sections.

5 Release a similar task. Pupils plan and complete a closely matched example without the support, then check against the success criteria.

What this sounds like

“The first three steps are complete. Finish the conclusion and explain how the evidence above supports it. On the next example, you will supply the final two steps.”

Classroom example. A Maths completion task supplies the setup and first lines but leaves the final decision blank. Pupils finish and explain the rule they used. The next task removes the final two stages while keeping the demand comparable. A clean independent question follows, checking whether pupils can organise the method without the visible structure.

Top tips that make the technique work

Start with the final step. Backward fading keeps the earlier structure available while students finish the process.

Keep demand comparable. Do not remove support and introduce a new representation or harder content in the same move.

Make the blank meaningful. A missing decision reveals more than an obvious word copied from the line above.

Ask for the principle. Students should explain why their completion fits the method or criterion.

Finish with a clean attempt. A complete independent example checks whether the visible structure was doing too much of the work.

Do and avoid

Do	Avoid
Increase responsibility in manageable stages.	Removing random lines without a rationale.
Keep the underlying task stable while fading.	Filling blanks that require no meaningful decision.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Fading Support | Barton - We Do

Fading scaffolds

Removing support as students become secure

Use fading to move students beyond dependence on prompts, models or frames. Decide which support to remove from evidence of what students can now do.

How to do it well

1 Identify the remaining dependency. Observe which part of the task still requires the model, cue or teacher prompt.

2 Select the next reduction. Remove one support that successful performance suggests is no longer needed.

3 Give a comparable attempt. Hold the task demand steady so you can interpret the effect of the reduction.

4 Respond to the attempt. Keep, reduce or briefly restore a specific cue according to accuracy and reasoning.

5 Hand over the process. Students plan and complete the task independently, then revisit it later and in a varied context.

What this sounds like

“You no longer need the sentence opening. Keep the evidence checklist for this attempt; next time we will try without that too.”

Classroom example. For an extended response, the teacher first removes the sentence starters but keeps the evidence checklist. After pupils can explain the links independently, the checklist is reduced too. When one student loses the structure, only the necessary cue returns. Support changes with the observed barrier rather than disappearing for the whole class at once.

Top tips that make the technique work

Separate support from difficulty. Keep content familiar while removing a cue, then introduce variation once independence is secure.

Fade a specific feature. Remove the sentence opening, checklist item or worked step that the student no longer needs.

Watch for hidden dependence. Successful copying from a visible model does not establish that the student can choose the approach alone.

Restore the minimum cue. If success drops, reinstate the smallest support that repairs the barrier.

Include later independence. Return after a delay so the task is not supported by immediate memory of the model.

Do and avoid

Do	Avoid
Make the withdrawal staged and purposeful.	Keeping a frame permanently for every task.
Check the decisions students make without prompts.	Removing all support on the same date for every student.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Fading Support | DistillED - Provide Scaffolds for Difficult Tasks

Fluency practice

Securing the essential steps

Use fluency practice when students understand a process but perform it slowly or inconsistently. Secure accuracy before increasing pace, volume or complexity.

How to do it well

1 Identify the component. Locate the decision or action that is accurate but hesitant, or repeatedly disrupts the whole task.

2 Set a short focused practice set. Keep the target clear and begin with familiar forms.

3 Inspect early attempts. Correct the faulty step before it is repeated and provide a small prompt or model where needed.

4 Increase efficiency. Reduce prompts and encourage smoother execution while preserving accuracy and explanation.

5 Recombine and revisit. Use the component in a complete task, then return after a delay and introduce useful variation.

What this sounds like

“For these six items, practise selecting the correct operation. Accuracy comes first. After that decision is secure, we will complete the full problems.”

Classroom example. In Music, students practise a short rhythmic component that interrupts a longer performance. The teacher first secures accuracy, then smoother execution and finally recombination with the full phrase. Later rehearsal revisits the same component. Faster playing is useful only when it preserves the timing and technique being practised.

Top tips that make the technique work

Practise the bottleneck. A short set on the difficult component may be more useful than repeatedly completing the whole task.

Keep correct models available initially. Students should not spend the practice interval repeatedly rehearsing an error.

Increase speed cautiously. Fluency includes accurate, efficient performance; rushing alone is not the aim.

Use purposeful variation. Vary features that require attention while keeping the target skill stable.

Include delayed practice. Smooth performance immediately after modelling can fade unless revisited.

Do and avoid

Do	Avoid
Monitor early items and correct promptly.	Timed competition before the method is secure.
Build towards accurate independent performance.	Repeating identical items without attending to their structure.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

Barton - Purposeful Practice | DistillED - Require and Monitor Independent Practice

Variation

Changing examples to deepen understanding

Use variation to help students notice the structure of an idea rather than copy surface features. Decide what changes, what stays the same and why that comparison matters.

How to do it well

1 Choose the relationship. Identify the feature, rule or structure students should notice across examples.

2 Establish a baseline. Solve or explain one accessible example so the comparison has a clear starting point.

3 Change a consequential feature. Keep other demands stable and ask students to predict the effect.

4 Compare the outcomes. Discuss what changed and what remained valid, using the underlying principle rather than appearance.

5 Test the generalisation. Present a less familiar case and ask students to justify whether the same relationship still applies.

What this sounds like

“Only the direction of the arrow has changed. Predict what that changes in the explanation, and identify what remains true.”

Classroom example. A Science teacher changes one condition in a familiar investigation and asks pupils to predict the consequence before calculating or explaining. Pupils compare the new outcome with the baseline and identify the relationship that remains true. A further example tests the generalisation, so variation directs attention to structure rather than mere novelty.

Top tips that make the technique work

Change deliberately. A sequence is not purposeful variation simply because every question looks different.

Include an unchanged relationship. Students need something stable against which to notice the new feature.

Ask for a prediction. Before calculation or completion, ask what they expect to change and why.

Use near-misses. A small change that makes a case invalid can clarify a concept boundary.

Compare after solving. Ask students to explain the effect of the change rather than moving straight to another item.

Do and avoid

Do	Avoid
Select changes that reveal the target relationship.	Changing several unfamiliar features at once.
Ask students to describe the invariant.	Assuming a varied worksheet automatically produces conceptual understanding.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

Barton - Purposeful Practice | DistillED - Non-Examples

Interleaving

Practising how to choose an approach

Use interleaving when students can perform several approaches separately but need practice deciding which one fits. Mix types that require a meaningful choice after initial teaching.

How to do it well

1 Check initial competence. Confirm that students can use each approach in a familiar form with reasonable accuracy.

2 Select contrasting types. Choose tasks for which students must discriminate between related methods or concepts.

3 Mix the sequence. Avoid blocks that reveal which method comes next, while keeping demands manageable.

4 Require the choice. Ask students to identify the approach and explain the clue before or alongside completing the response.

5 Give comparative feedback. Address why an alternative did not fit, then revisit the choice in later mixed practice.

What this sounds like

“These questions need different methods. Before calculating, name the method you would choose and the feature in the question that tells you.”

Classroom example. A Maths practice set mixes tasks requiring different familiar methods. Pupils identify the method and the clue before solving. Feedback compares a plausible alternative with the chosen approach. The difficulty lies partly in selection, whereas a worksheet labelled with one method would have supplied that decision before the pupil began.

Top tips that make the technique work

Teach each approach first. Interleaving should not disguise missing instruction as desirable difficulty.

Remove the topic label. A heading such as “Use this formula” can supply the decision you want students to practise.

Ask for classification before execution. Students can state the approach and the clue that makes it suitable before completing the task.

Mix related alternatives. Contrast methods students might plausibly confuse, rather than assembling a random set.

Expect performance to feel harder. Inspect the nature of errors and provide feedback; difficulty alone is not evidence the sequence is effective.

Do and avoid

Do	Avoid
Practise selection as well as execution.	Mixing untaught content into independent practice.
Keep the approaches individually accessible.	Giving a method cue above every item.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Research Spacing and Interleaving

Independent practice

Helping students begin and sustain effort

Use independent practice to build accurate, sustained performance after explanation and guided rehearsal. Plan the launch, first items and monitoring so students remain thinking throughout.

How to do it well

1 Prepare for success. Use evidence from guided practice to confirm that most pupils can complete the process accurately and identify who still needs a scaffold or teacher check.

2 Rehearse before release. Correct common misconceptions, make the success criteria visible and clarify what pupils should do if they become stuck.

3 Mirror before you leap. Begin with the same method, representation and type of material used in guided practice. Do not hide new knowledge inside an 'independent' task.

4 Monitor in motion. Circulate from the first item, sample the whole room and intervene before an error is repeated across the set.

5 Build fluency through repetition. Secure accuracy first, then reduce hesitation, increase volume or add carefully selected variation.

6 Connect to future retrieval. Return to the learning after a delay and mix it with related content so performance becomes durable and flexible.

What this sounds like

“Begin with the first two items independently. If stuck, identify the step that is unclear and use the model. When finished, explain why your method works.”

Classroom example. A teacher begins independent practice with two familiar items, clear success criteria and a taught stuck protocol. A short observation establishes the working norm before circulation. Early answers reveal one shared error, so the class receives a bounded repair. Pupils then return to productive practice with a meaningful explanation task for early finishers.

Top tips that make the technique work

Protect the first moments. A brief Golden Silence establishes the expectation before circulation begins.

Teach what to do when stuck. Students can check the model, identify the uncertain step or move to a permitted item; repeated hand-up waiting reduces participation.

Inspect the first attempt. Early diagnosis prevents a whole page of repeated errors.

Provide a meaningful finish task. Deepen the same learning through explanation or variation rather than unrelated extra work.

Preserve productive time. Interrupt only when a common issue makes the class pause worthwhile.

Do and avoid

Do	Avoid
Check readiness before release.	Hiding new learning inside the independent task.
Use a route that includes the whole room.	Equating a quiet classroom with sustained thinking.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Require and Monitor Independent Practice | Barton - Participation during independent practice

Wait time

Giving everyone time to think

Use wait time to give everyone a chance to retrieve, reason and prepare an answer. Include a pause after a student response before immediately evaluating or asking someone else.

How to do it well

1 Plan a space-worthy prompt. Use a question that needs retrieval, explanation, justification or evaluation; the required pause should match its complexity.

2 Set the expectation. Explain that everyone will think before anyone answers and that jotting may be used when the response has several parts.

3 Wait Time 1: pause after the question. Ask the complete question, then silently count at least three to five seconds—or longer—without rephrasing, filling the silence or signalling a respondent.

4 Sample an answer. Cold call or use the planned response routine so the protected thinking time belongs to the whole class, not only volunteers.

5 Wait Time 2: pause after the response. Let the answer land before evaluating it; give peers time to process, then invite elaboration, a build or a challenge.

What this sounds like

“Everyone think silently about which source is more reliable. I will give you time before asking for an answer. Prepare the reason as well as your choice.”

Classroom example. After asking pupils which source better supports a judgement, the teacher gives silent preparation time before naming a respondent. After the answer, another pause allows the class to consider the evidence. A second pupil then compares the reasoning. The question gives more students access than an immediate request for the quickest raised hand.

Top tips that make the technique work

Match the pause to the question. A complex explanation requires more preparation than a brief factual response; no single number fits every prompt.

Keep the name until after thinking. Naming a pupil first can signal that others need not prepare.

Avoid filling the silence. Rephrasing continuously interrupts the thinking you intended to protect.

Use a jot where useful. A short written commitment gives students a stable starting point for a longer response.

Preserve the second pause. After an answer, allow others to consider it before asking for agreement, challenge or refinement.

Do and avoid

Do	Avoid
Make protected thinking time predictable.	Treating the fastest answer as the best answer.
Keep every student accountable for the prompt.	Announcing thinking time and immediately supplying a hint.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Wait Time | Barton Cold Call

Cold calling

Involving everyone in questioning

Use cold calling to make classroom questions an invitation for everyone to think. Build a warm, predictable question–pause–name routine and use answers to guide teaching.

How to do it well

1 Establish the rationale and norms. Explain that everyone thinks, the teacher selects and uncertainty is a normal starting point—not a public trap.

2 Ask the whole class. Pose a clear factual, conceptual or reasoning question before naming a pupil.

3 Protect wait time. Allow at least several seconds, and longer for complex prompts; scan the room while everyone prepares.

4 Invite warmly. Name the pupil with open, supportive language and listen without interrupting the first attempt.

5 Probe or connect. Ask for evidence, a clearer term, a connection or another pupil's build; require the class to listen.

6 Support successful re-entry. Use paired or written rehearsal, pre-call, a prompt, a second pupil's answer and a return to the original pupil when confidence or knowledge is fragile.

What this sounds like

"Without hands up or calling out, everyone think: which evidence supports that claim? [Pause.] Priya, what did you choose and why?"

Classroom example. A teacher asks everyone to consider why a line of code fails, pauses, then names a pupil. An incomplete response receives a targeted clue and preparation time before the teacher returns. The routine keeps the public answer supportive while requiring everyone to think; it is not used to catch out a pupil who was distracted.

Top tips that make the technique work

Front-load the response mode. Tell students to think without calling out before posing the question.

Ask before naming. Everyone prepares, then you select the respondent.

Use preparation strategically. Silent thinking, a jot or partner rehearsal helps students develop a response before public contribution.

Distribute questions deliberately. Sample across the room, avoiding repeated reliance on confident pupils or random selection without a purpose.

Respond to uncertainty constructively. Prompt, provide rehearsal or a Second Chance; do not turn a difficult answer into public punishment.

Do and avoid

Do	Avoid
Keep the routine predictable and supportive.	Using a question as a sanction for talking.
Ask follow-up questions about the reasoning.	Naming a pupil before others have had time to think.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Cold Calling | Barton - Cold Call

Think Pair Share

Preparing everyone to answer

Use Think–Pair–Share when students need independent thinking and private rehearsal before a public response. Protect the Think stage so the quickest speaker does not supply everyone's first idea.

How to do it well

1 Design a thinking-worthy prompt. Ask pupils to retrieve, connect, compare, justify or explain; keep the prompt visible as an anchor.

2 Protect THINK. Give uninterrupted silent time for each pupil to retrieve and, when useful, jot keywords, evidence or a first decision.

3 Structure PAIR. Specify who speaks first, what the listener must do, when roles swap and what the pair must improve or agree.

4 Run an inclusive SHARE. Cold call pairs or individuals so either partner may summarise, justify or report the other person's idea; probe contrasting reasoning.

5 Adapt the sequence. Use Think–Write–Pair–Share, partner-summary writing, mini-whiteboards or a batched share when these strengthen retrieval or accountability.

What this sounds like

"Think alone and write one reason. Partner A explains first; Partner B checks the evidence. Swap, then be ready to share your partner's strongest point."

Classroom example. In Geography, pupils individually choose the most important cause and write a reason. Partners compare evidence with clear speaking roles, then pupils share a partner's strongest point. A final individual explanation captures the refined thinking. The initial silent stage prevents one confident partner supplying both pupils' starting ideas.

Top tips that make the technique work

Make thinking silent. Ask for a mental answer or short jot before anyone turns to a partner.

Choose a worthwhile prompt. Explanation, comparison and justification give partners something to develop.

Specify the first speaker. A simple A/B or seating rule prevents negotiation and domination.

Give the listener a task. Restate, question or identify the strongest reason rather than wait for a turn.

Share through accountable sampling. Cold call or collect a fresh individual response; do not assume hearing lively talk means everyone understood.

Do and avoid

Do	Avoid
Keep all three stages distinct.	Starting the conversation before individual thinking.
Require both partners to contribute.	Allowing the Share stage to return to volunteers only.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Think-Pair-Share | Barton - Turn and Talk

Turn and Talk

Structuring purposeful partner discussion

Use Turn and Talk for a short, purposeful exchange that helps students clarify or rehearse an idea. Specify the question, partner roles, time and response you will collect afterwards.

How to do it well

1 Build predictable pairs. Use stable A/B partners where possible and decide the speaking order, including a planned trio when numbers require it.

2 Give thinking time. Protect five to ten seconds—or longer—so each pupil retrieves, annotates or drafts before speaking.

3 Structure the question. Ask pupils to explain, justify, compare, predict, evaluate or rank, using content already taught or made available.

4 Provide a concrete stimulus. Anchor talk in a text, graph, model, image, worked example, data or vocabulary prompt rather than memory-free opinion.

5 Rehearse briefly and account. Use short bursts, switch speakers explicitly, stop near the crest of productive talk and cold call pupils to share or summarise.

What this sounds like

“Partner A explains which option fits; Partner B challenges the reason. Swap after thirty seconds. Afterwards I may ask you to explain your partner’s choice.”

Classroom example. In English, paired talk compares two interpretations of a line. Each partner must give a reason and respond to the other. The teacher samples the discussion, signals its end while useful talk is active and asks selected pupils to explain their partner’s reasoning. Students then refine their own written interpretation.

Top tips that make the technique work

Prime the end as well as the start. Students should know how they will stop and what answer they must be ready to give.

Stop at the crest of the wave. End while useful exchange is active rather than waiting until every pair has exhausted the prompt.

Listen for the thinking. Sample the content of talk, not merely whether pupils are speaking.

Ask about the partner. A request to restate the other person’s reasoning creates listening accountability.

Re-vote individually where appropriate. After comparing ideas, students commit again so you can see changed thinking.

Do and avoid

Do	Avoid
Set a specific cognitive question and clear roles.	“Discuss this” with no product or purpose.
Harvest the discussion deliberately.	Letting one partner deliver both contributions.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Turn and Talk | Barton - Turn and Talk

Choral response

Rehearsing brief answers together

Use choral response for brief, unambiguous answers that benefit from shared rehearsal. Teach a precise cue so students begin together rather than copying the earliest voice.

How to do it well

1 Choose suitable content. Use choral response for a brief fact, phrase, pronunciation, definition or completed step with one clear answer—not an extended explanation.

2 State and model the routine. Teach the cue, expected volume and complete response; model difficult vocabulary before the class attempts it.

3 Ask, think, cue. Pose the prompt, allow silent retrieval and give one unambiguous signal so every voice begins together.

4 Listen for participation and accuracy. Attend to timing, confidence and the content of the response; reset if voices are missing or the response is ragged.

5 Affirm or correct immediately. Name what was accurate, or provide the correct answer with a brief explanation and have the class repeat it precisely.

6 Check individuals when needed. Follow with cold call, mini-whiteboards or a process question; a chorus cannot by itself prove each pupil understands.

What this sounds like

“Think first. On ‘everyone in three, two, one’, say the missing word together. Plants take in carbon… Everyone in three, two, one.”

Classroom example. A Languages teacher rehearses a short phrase on a common cue after protected thinking time. The teacher listens for accurate pronunciation and repeats the troublesome part. A later individual written response checks the meaning. The chorus supports shared rehearsal without being treated as evidence that every pupil independently understands the phrase.

Top tips that make the technique work

Choose a bounded response. A key term, short fact or agreed phrase works better than an open explanation.

Build thinking before the cue. Students should know the answer they intend to give before the chorus starts.

Rehearse with easy content. Establish the mechanics separately from challenging subject knowledge.

Listen for accuracy. A loud chorus can hide missing, delayed or incorrect responses.

Use another check for diagnosis. When you need individual evidence, follow with a board response, written answer or targeted sample.

Do and avoid

Do	Avoid
Use a consistent, rehearsed response cue.	Treating volume as proof of individual understanding.
Keep answers short and precise.	Using an ambiguous prompt with several possible responses.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Choral Response | Barton - Call and Respond

Probing questions

Moving beyond the first answer

Use probing questions after students have relevant knowledge to deepen or clarify the reasoning behind an answer. Plan a focused follow-up rather than asking a vague “Why?” every time.

How to do it well

1 Establish the knowledge. Explain or model the method, concept or example before asking pupils to reveal how it works.

2 Anchor the probe. Begin from a pupil answer, a worked step, a comparison or a decision the class can see.

3 Ask one focused how or why. Target the clue used, the first step, the reason for a choice, a link to prior knowledge or an error that must be corrected.

4 Refine and rehearse. Prompt the pupil to add evidence, use precise vocabulary, sequence the explanation or improve it after hearing a peer.

5 Return to the process. Revisit the reasoning in a related example so the method becomes transferable rather than a one-off performance.

What this sounds like

“You chose the second source. Which feature makes it more trustworthy for this question? What evidence would weaken your judgement?”

Classroom example. A pupil identifies the correct result but offers little reasoning. The teacher asks which feature ruled out the alternative, then asks the pupil to complete the explanation using the key term. Another pupil compares the reasoning with a second case. The follow-up deepens a specific decision instead of becoming a vague extended interrogation.

Top tips that make the technique work

Probe a consequential decision. Ask which clue, criterion or relationship produced the answer.

Plan likely follow-ups. Prepare a question for an incomplete answer, a misconception and a secure explanation.

Hold the pupil’s thinking. Give a prompt, then return to the same student to complete the response.

Use precise language. “Which feature ruled out A?” is often clearer than “Tell me more.”

Involve listeners. Ask others to restate, compare or challenge the reasoning before moving on.

Do and avoid

Do	Avoid
Anchor the probe to a visible answer or step.	Asking students to reason about untaught material.
Prompt a fuller disciplinary explanation.	Finishing their answer for them and accepting your own explanation.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Process Questions | Barton - Plan follow-up questions in advance

Compare justify evaluate

Questions that deepen reasoning

Use comparison, justification and evaluation prompts when students need to explain a relationship or make a judgement. Ensure they have the knowledge and criteria required to reason well.

How to do it well

1 Choose the thinking demand. Decide whether students are identifying a contrast, defending a claim or judging quality against a criterion.

2 Prepare the cases. Select examples close enough to require attention to the important difference.

3 State the criteria. Clarify which knowledge, evidence or performance feature should inform the response.

4 Protect preparation. Give silent time to form a claim and supporting reason, then use talk or writing to refine it.

5 Probe and transfer. Ask what would change the judgement and test the same criterion in a fresh case.

What this sounds like

“Compare the two methods for this task. Which is more efficient, and what evidence supports that judgement? When might the other method be preferable?”

Classroom example. In Art, pupils compare two compositions against a taught criterion. They identify a feature, explain its effect and justify which better meets the purpose. A changed brief then asks whether the judgement would change. The evaluation draws on disciplinary knowledge rather than a preference for whichever work initially looks more attractive.

Top tips that make the technique work

Make comparison focused. Specify the feature or relationship to compare rather than asking for every possible similarity and difference.

Require evidence. “I prefer this one” becomes useful when linked to a criterion or example.

Choose genuinely comparable cases. If one is obviously poor, the judgement may demand little thought.

Ask for an alternative. Students can explain what would make the other option defensible.

Rehearse before public answers. A written claim and evidence or structured partner exchange can improve the precision of the response.

Do and avoid

Do	Avoid
State the criterion and demand a reason.	Vague preferences without disciplinary evidence.
Use taught knowledge to support the judgement.	An evaluation task before students know what quality means.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

DistillED - Process Questions | Barton - Plan follow-up questions in advance

Mini-whiteboards

Checking everyone's understanding

Make every student’s answer visible, identify gaps and decide what to teach next. Useful before independent practice or whenever answers from a few students leave you unsure about everyone else.

How to do it well

1 Plan a revealing question. Target one idea, anticipate plausible wrong answers and choose a word, calculation, sketch or choice that can be scanned.

2 Rehearse the mechanics. Model clear writing, private answers, boards facing you and erasing only on signal; practise with an easy question.

3 Secure independent thinking. Give response instructions before the question, allow time and keep boards hidden until commitment is complete.

4 Collect every response. Reveal together and scan systematically. If revealing back to front, everyone stops writing before the first row shows.

5 Probe and respond. Look for common errors and missing responses, ask how selected students decided, then Pause & Fix, Adapt & Support or Extend & Challenge as the evidence warrants.

What this sounds like

“One formula, written large. Keep it hidden. Write the formula for magnesium chloride.” [Thinking time.] “Pens down. Show me in three, two, one. Hold still while I check.”

Classroom example. Several Chemistry boards show MgCl rather than $MgCl_2$. The teacher keeps the responses private until all pupils commit, identifies the pattern and re-models charge balance. Pupils then answer a different ionic formula question. Correcting the original board alone would have shown compliance with the model rather than independent application.

Top tips that make the technique work

Separate working from the answer. Use one side for working and the other for a large final answer; turn it over when explaining the method.

Bring bookwork onto boards. Transfer the answer to one pivotal book question onto a board, then scan before showing the solution.

Investigate blank boards. Ask what the student knows or which part is unclear; offer a starting prompt and return.

Control public answers. Avoid praising an early correct board while others are writing; gather the evidence before revealing the solution.

Keep the routine available. Working pens, erasers and boards should be within reach; re-model premature erasing or unreadable answers.

Do and avoid

Do	Avoid
Scan every response before giving the answer.	Accepting the first correct board as evidence for everyone.
Probe reasoning and use a fresh check after repair.	Using boards to rank students or as punishment.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED Mini-whiteboards revisited | Barton Top five tips

Barton Detailed routines | Barton Back row first

Hinge questions

Deciding whether to move on

Use a hinge question at a decision point where students' answers should change what happens next. Plan the question and the teaching response together.

How to do it well

1 Plan the hinge in advance. Choose a pivotal concept and build plausible distractors from common misconceptions rather than random errors.

2 Ask it after a small teaching chunk. Place the question just before the lesson would otherwise move into the next explanation or practice demand.

3 Protect independent commitment. Give silent thinking time, keep the answer hidden and collect a whole-class response before partner discussion where initial diagnosis matters.

4 Scan choices and reasoning. Look for patterns, then invite pupils to explain why an option was attractive before revealing the correct answer.

5 Make the instructional decision. Move on when understanding is sufficiently secure; otherwise re-explain the identified misconception, use a different example or extend guided practice.

6 Recheck and refine. Test the same concept in a related form and later improve distractors that did not reveal the intended thinking.

What this sounds like

"Choose privately and prepare your reason. We will use the responses to decide whether we are ready for the next example or need to revisit this distinction."

Classroom example. Before a new stage in a calculation, a hinge question checks the relationship needed for that stage. Each option represents a different likely misconception. Independent choices reveal a shared error, so the teacher pauses, contrasts the methods and asks a fresh question. The planned next task changes because of the evidence.

Top tips that make the technique work

Identify the hinge. Choose a concept that students need before the next task or explanation will make sense.

Match each distractor to an error. An option should tell you what needs clarifying, not just whether the answer is wrong.

Collect independent answers first. Discussion can improve learning afterwards, but initial private commitment gives cleaner diagnostic evidence.

Decide your likely responses. Plan what you will do for a common misconception, mixed understanding and broadly secure answers.

Ask a fresh follow-up. The response after repair should require application, not memorisation of the original option.

Do and avoid

Do	Avoid
Place the question before a consequential increase in demand.	A quiz item with no effect on the next teaching move.
Inspect reasoning and the pattern of choices.	Automatically moving on when a fixed percentage is correct.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Hinge Questions | Barton - Diagnostic questions

Checking for understanding during explanations

Use checks during and between explanation chunks to find out what students can recall, distinguish or apply. Gather evidence that informs a specific teaching decision.

How to do it well

1 Teach it first. Explain or model one coherent step before asking pupils to demonstrate understanding; do not use a cold question to test untaught material.

2 Ask a specific question. Ask what pupils understand, can distinguish or can do—not whether they understand. Script high-value recall, application, misconception and process prompts in advance.

3 Pair-share after silent thinking. Give 5–10 seconds to think, then a brief, structured Turn and Talk with labelled partners and an expectation of full-sentence rehearsal.

4 Pick non-volunteers. Cold call after the rehearsal so everyone remains accountable and the public sample is not limited to confident volunteers.

5 Listen to the response. Attend to target vocabulary, reasoning and the pattern across responses. Probe incomplete thinking instead of fishing for the answer.

6 Give effective feedback and act. Echo a secure answer, elaborate tentative thinking, or explain and reteach an error. Aim for about 80% success before moving on, then continue support until everyone can access the next step.

What this sounds like

“First answer alone. Now compare the reasoning with your partner. I will ask for explanations, then everyone will try a new example independently.”

Classroom example. After a short explanation, students first answer a new case alone. They compare reasoning in pairs, and the teacher samples non-volunteers before a fresh individual check. Responses show whether the difficulty concerns the principle or a missed instruction. The teacher can then choose a specific repair rather than repeat the whole explanation.

Top tips that make the technique work

Distinguish listening from understanding. Repeating what was said checks receipt; a fresh application checks whether the idea can be used.

Plan five useful checkpoints. Do Now review, prerequisites, basics after explanation, hinges and readiness for independent practice offer different evidence.

Use TAPPLE thoughtfully. Teach, Ask, Pair-share after independent thinking, Pick non-volunteers, Listen, and give Effective feedback.

Preserve independent evidence. Collect an initial answer before partner talk where you need to diagnose each pupil’s starting understanding.

Prepare the response. A check is useful when it helps you decide whether to clarify, practise, support or extend.

Do and avoid

Do	Avoid
Check the specific idea just taught.	“Any questions?” as the main evidence.
Respond and use a fresh related question.	Treating a correct volunteer answer as class readiness.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Check for Student Understanding | Barton - Checking for understanding Ten key principles

Barton Responsive teaching

Circulation

Sampling independent work and responding

Use circulation to read learning across the room, identify the exact error and choose a response. Move with a route and a diagnostic focus rather than following raised hands alone.

How to do it well

1 Plan the route. Identify the success criterion, likely errors and pupils you must see, with a path that covers every area.

2 Observe first attempts. After securing the launch, inspect early work before students repeat the process many times.

3 Diagnose the response. Read the relevant section and ask a brief question where the reasoning is unclear.

4 Give a bounded next action. Use a prompt, clue or direct explanation according to need, then require the pupil to do the thinking.

5 Revisit and decide. Check the change; preserve productive work where gaps are isolated, or pause the class when the pattern justifies reteaching.

What this sounds like

“I am checking the reason you chose this method. Show me that decision. Fix this line, then try the next item; I will return shortly.”

Classroom example. During essay practice, the teacher’s route checks the claim–evidence link in selected first paragraphs across the room. Quick notes distinguish missing evidence from weak explanation. Some pupils receive an individual prompt; a shared pattern becomes a brief class model. The teacher returns to inspect the changed sentence before continuing the route.

Top tips that make the technique work

Inspect one feature at a time. Look for the first decision, selected evidence or a specific presentation feature so the scan stays purposeful.

Include likely invisible pupils. Deliberately visit students whose work or thinking you do not normally hear.

Visit strong work early too. It helps establish the expected standard and can supply an exemplar for later discussion.

Keep the whole room visible. Step back between individual interactions; avoid becoming absorbed at one desk.

Record patterns briefly. A quick note or seating-plan mark helps distinguish an individual gap from a class-wide issue.

Do and avoid

Do	Avoid
Use scan, read and compare with the criterion.	Wandering without a focus.
Return after a targeted intervention.	Explaining at length beside one pupil while losing the rest of the room.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Circulation and Actionable Feedback | Barton - Participation during independent practice

Exit tickets

Identifying what needs revisiting

Use an exit ticket when a brief independent response can inform the next lesson. Choose essential knowledge and a format that can be reviewed quickly.

How to do it well

1 Choose the purpose. Identify the essential idea or prerequisite you need to inspect before the next lesson.

2 Write a short prompt. Use a bounded explanation, diagnostic choice, diagram or small performance that reveals the target knowledge.

3 Allow an independent response. Reserve enough time and keep the solution hidden until students have committed.

4 Divide and inspect. Sort by quality of understanding, then look closely at the main error patterns and selected unclear answers.

5 Decide the next lesson response. Plan a fresh retrieval or diagnostic prompt and the explanation, rehearsal or support the evidence warrants.

What this sounds like

“Before you leave, answer this alone: which example shows diffusion, and which feature tells you? Include one sentence explaining your choice.”

Classroom example. A Computing exit ticket asks pupils to predict the output of a short, taught code structure and explain one decision. The teacher sorts the answers and finds a common misunderstanding. The next lesson begins with a related check and a short repair, rather than starting new content over an unresolved prerequisite.

Top tips that make the technique work

Keep the ticket narrow. One useful conceptual distinction may reveal more than five broad questions.

Include reasoning where necessary. A letter choice alone can hide why the student selected it.

Protect independence. A ticket copied after class discussion gives less information about the individual’s understanding.

Sort before annotating. Group responses by secure, partly secure and insecure, then inspect the causes of difficulty.

Prepare the next move. Decide whether tomorrow needs retrieval, a brief clarification, guided practice or a fuller reteach.

Do and avoid

Do	Avoid
Target the learning that matters for the next lesson.	Collecting tickets without reviewing them.
Use the responses to plan teaching.	A long end-of-lesson assessment with no time to complete it.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Exit Tickets | Barton - Diagnostic questions as exit tickets

Pause and Fix

Responding to a common misconception

Use Pause & Fix when a common gap means continuing would reinforce an error or make the next step inaccessible. Stop briefly, repair the precise issue and return to a fresh attempt.

How to do it well

1 Identify the common gap. Use responses or work to locate the knowledge or decision causing widespread difficulty.

2 Stop and frame the repair. Secure attention, name the issue neutrally and explain why this short pause matters.

3 Re-model differently. Use a smaller step, contrasting example or clearer representation that addresses the error directly.

4 Give a fresh check. Require every student to apply the idea to a related example before you reveal the answer.

5 Resume proportionately. Continue when the evidence supports it, with targeted follow-up for remaining gaps; extend the repair when the class is still insecure.

What this sounds like

“Pause. Several answers use the mass in place of the volume. We will model that distinction once, then everyone will try a new example.”

Classroom example. In Maths practice, several pupils apply the same invalid cancellation. The teacher stops, identifies the exact condition for a valid move and contrasts the two cases. Everyone then attempts a new item. Once the class pattern is secure, practice resumes with individual follow-up for remaining gaps rather than a complete restart of the topic.

Top tips that make the technique work

Pause for a consequential pattern. A few isolated slips may need individual feedback rather than stopping productive class work.

Change the explanation. Repeating the same long account more loudly rarely addresses the faulty idea.

Keep the repair bounded. State the specific distinction you are fixing and show it with one clear example.

Keep secure students thinking. Ask them to explain the misconception or test a useful variation rather than sit through unnecessary repetition.

Recheck before resuming. Use changed surface details and the same conceptual demand so students must apply the repair.

Do and avoid

Do	Avoid
Match the repair to the shared error.	Stopping the class for every interesting error.
Secure attention before re-explaining.	Treating copied corrections as proof the issue is fixed.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - A model for responsive teaching | DistillED - Adaptive Teaching

Adapt and Support Extend and Challenge

Responding to different needs

Use Adapt & Support and Extend & Challenge when responses show different needs. Keep the learning goal ambitious while adjusting the support or depth of the next task.

How to do it well

1 Read the pattern. Identify which students need a prompt, a re-model or more guided practice and which are ready for deeper application.

2 Locate the barrier. Check whether the issue concerns knowledge, vocabulary, task structure or a specific decision.

3 Select the smallest support. Give a useful cue, model, frame or short teacher group while retaining the curricular goal.

4 Extend the secure learning. Use explanation, variation or a less familiar case that requires additional thinking.

5 Hand back responsibility. Remove support as competence becomes visible and use independent responses to decide the next adjustment.

What this sounds like

“If you need the first-step prompt, use it for this example. If your method is secure, explain how the answer changes when this condition changes.”

Classroom example. In Science, some pupils need a prompt to select the relationship while others can justify it independently. A short teacher group rehearses the first decision; secure pupils explore how a changed condition affects the result. All work remains connected to the same concept, and the temporary support is removed when students can proceed alone.

Top tips that make the technique work

Diagnose before grouping. The same incorrect answer can come from different barriers; a quick probe may change the support needed.

Use flexible groups. Group briefly by the current learning need, then release students when they can proceed independently.

Extend through depth. Ask for justification, a boundary case or transfer rather than simply more identical questions.

Preserve a common thread. Students should remain working on the intended knowledge even when support differs.

Return to independent evidence. A successful response beside the teacher may depend on prompts; check what the student can do alone.

Do and avoid

Do	Avoid
Adjust support to the observed barrier.	Permanent task assignment by label or prior attainment.
Provide meaningful challenge for secure students.	Giving different groups unrelated learning goals.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Adaptive Teaching | Barton - A model for responsive teaching

Actionable feedback

Giving students a useful next step

Use individual feedback when a precise next action can improve work while the thinking is still live. Make the location, learning issue and required change clear.

How to do it well

1 Plan and break the plane early. Name the success criteria, must-check pupils and a predictable route before work begins.

2 Cover the full field. Move systematically, stay aware of the room and avoid reactive zig-zagging or repeatedly visiting the same pupils.

3 Diagnose before feeding back. Scan, read and compare with the criteria; note exemplars, partial attempts, misconceptions and pupils to revisit.

4 Use prompt, clue, then direct. Start with the least support likely to activate the pupil's knowledge; escalate only when the response shows it is needed.

5 Bound the action. Identify the exact location and a finite improvement that the pupil can complete now.

6 Return and respond. Check the change, praise against criteria and pause for whole-class correction when the same misconception is widespread.

What this sounds like

“Your claim is clear. In this sentence, add evidence that supports it. Tell me what you will change, then make the edit and I will return.”

Classroom example. In PE, a pupil receives feedback on one feature of a taught movement. The teacher identifies what is already successful, gives a precise next action and asks for another attempt. A later return checks the change. The feedback is short enough to use immediately and does not become a list of every possible improvement.

Top tips that make the technique work

Begin with the successful feature. Precise recognition tells the pupil what to retain while improving another part.

Use prompt, clue, direct. Start with a question if the pupil has the knowledge; provide more guidance when the response shows it is needed.

Bound the action. One sentence or process step is easier to act on than “improve your explanation”.

Ask for immediate work. A feedback conversation should end with the student thinking, writing, rehearsing or trying again.

Return later. Check the change and an additional attempt rather than assuming the comment was understood.

Do and avoid

Do	Avoid
Name the exact feature and next action.	Vague praise followed by “more detail”.
Check listening and require a response.	Giving the answer when a usable prompt would suffice.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Circulation and Actionable Feedback | DistillED - Feedback as Detective Work

Whole-class feedback

Responding to patterns in work

Use whole-class feedback when students' work reveals shared strengths and recurring difficulties. Review the pattern once, then give everyone a useful improvement task.

How to do it well

1 Review student work. Read the relevant body of work to identify common strengths, misconceptions and repeated errors; avoid annotating every piece in detail.

2 Record key feedback. Use a crib sheet for success criteria, what went well, model answers, misconceptions, key vocabulary, SPaG patterns and next actions.

3 Plan the response. Choose a model, comparison, live edit, brief reteach, discussion or practice task that will turn the pattern into learning.

4 Deliver feedback visibly. Use the board, visualiser, screen or handout to show strong examples and typical errors without exposing or embarrassing pupils.

5 Get pupils thinking and improving. Allocate time for a focused response: redraft, rehearse, revisit, re-learn or research, according to the learning need.

6 Reflect and respond. Check the improvements, decide what requires future teaching and revisit fragile learning through practice, retrieval or a hinge question.

What this sounds like

"This feature was secure across many responses. This next error appeared often, so we will model it, try a related question and then improve our own work."

Classroom example. An English assessment reveals a recurring weak explanation after evidence. The teacher selects a model and near-miss, improves the connection live and asks pupils to complete a fresh equivalent paragraph. Work is then returned for an individual improvement task. Feedback becomes instruction and practice rather than repeated comments on many scripts.

Top tips that make the technique work

Select a small number of high-value issues. Two or three important misconceptions can be more useful than reviewing every mark lost.

Prepare before returning scores. Deliver the model and recheck while attention is on learning rather than individual results.

Use barometer work carefully. Selected work can reveal patterns, but do not assume a small sample captures every need.

Model, recheck, stretch. Use a fresh equivalent question after the explanation, then a useful variation when secure.

Make errors retrievable. Review Cards turn selected individual gaps into future practice, beyond today's correction.

Do and avoid

Do	Avoid
Choose the response from reviewed student work.	A presentation of comments with no improvement task.
Allocate time for students to act.	Reteaching every error for the same amount of time.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Whole Class Feedback | Barton - Whole-Class Feedback

Self-checking

Teaching students to improve their own work

Use self-checking when students know what quality looks like but do not yet inspect their own work reliably. Teach a specific checking action rather than telling them to “check it”.

How to do it well

1 Select the checking focus. Choose a consequential feature students have been taught and can recognise.

2 Demonstrate the check. Think aloud while inspecting a model and identifying a near-miss.

3 Rehearse on shared work. Students locate the feature or error in an example before moving to their own response.

4 Apply and improve. Require a visible checking action and an explained correction, rather than silent rereading alone.

5 Transfer the routine. Use the same checking question on a fresh task, then reduce the external prompt when students can apply it reliably.

What this sounds like

“Underline your claim and circle the evidence. Draw an arrow where the sentence explains how the evidence supports the claim. Which link is missing?”

Classroom example. In Science, pupils check the units in their calculations using a taught sequence: identify the quantity, inspect the substitution and inspect the final unit. They correct and explain a mismatch, then use the routine on another question. A generic instruction to check the whole page would have provided less guidance about the thinking required.

Top tips that make the technique work

Teach what to inspect. Specify a claim–evidence link, sign, unit or process decision rather than an unlimited search.

Use a model and a near-miss. Students need to see the feature present and absent before finding it in their own work.

Separate checking from rereading. A purposeful check asks a question of the work and may require recalculation, comparison or evidence tracing.

Give one checking focus first. Build fluency before expecting a long checklist to be used independently.

Check the checker. Sample how students located and repaired the issue; a tick beside a criterion is not evidence of accurate self-assessment.

Do and avoid

Do	Avoid
Model the checking process and require a visible action.	“Check your work” with no taught process.
Fade prompts as checking becomes reliable.	Assuming every self-awarded tick is accurate.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

DistillED - Feedback as Detective Work

Peer critique

Using criteria to improve work

Use peer critique when students understand the content and criteria well enough to offer accurate, useful feedback. Teach the feedback routine before exchanging work.

How to do it well

1 Establish the culture. Explain that critique is a learning tool and agree the norm: kind, specific and helpful.

2 Make quality visible. Clarify the criteria and compare model, near-miss and weak feedback comments.

3 Model SPARK live. Use anonymous work to demonstrate how a precise observation becomes a referenced, actionable suggestion.

4 Rehearse before exchange. Co-construct comments, evaluate examples and use sentence stems until pupils can distinguish strong from vague feedback.

5 Give and receive feedback. Begin with a genuine strength, then identify one high-impact improvement connected to the criteria.

6 Act and close the loop. The recipient decides how to use the comment, revises the work and explains the change; the teacher samples quality and accuracy.

What this sounds like

“Identify one success against the criterion. Then pinpoint one place to improve and suggest an action. The writer will decide how to use it and explain the change.”

Classroom example. In Art, pupils first compare useful and vague critique against one criterion. They then give a precise observation and an actionable suggestion on a partner’s work. The recipient evaluates the suggestion, revises the selected feature and explains the change. The teacher samples whether comments are accurate as well as respectful.

Top tips that make the technique work

Use SPARK. Comments should be Specific, Prescriptive, Actionable, Referenced to a criterion and Kind.

Model a weak comment too. Comparing “good work” with a precise suggestion helps students see what usefulness means.

Start with one criterion. Narrow the demand before expecting peers to judge an entire piece.

Keep responsibility with the recipient. Students should evaluate the comment, make a justified change and explain it.

Sample accuracy. Peer feedback does not replace teacher expertise when a misconception is difficult to diagnose.

Do and avoid

Do	Avoid
Teach the criteria and model useful comments.	Exchanging work before students know the standard.
Require the recipient to act.	Personal preferences or vague praise presented as critique.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Peer Feedback

Feedback tasks

Planning time to act on feedback

Use feedback tasks to make improvement an act of thinking. Select a solvable challenge that helps students locate, explain and repair an issue rather than simply copy a correction.

How to do it well

1 Choose a narrow learning focus. Identify the knowledge, skill or misconception that matters and that pupils can realistically improve.

2 Record what needs fixing without annotating it. Mark or review the work, then state the number and type of issues rather than correcting each location.

3 Pose a solvable search. Tell pupils what they are looking for and what success will look like, but leave the locating and correcting to them.

4 Add the minimum cue if needed. Narrow the section, paragraph, step or error type without revealing the exact line or answer.

5 Protect in-class time to act. Return the work when pupils have time to search, think, revise and use relevant models or criteria.

6 Close the loop. Ask pupils to explain what they changed and why, check the correction and later test whether they can apply the learning independently.

What this sounds like

“Two statements misuse the key term. Find them, explain the issue and correct them using the definition and the model.”

Classroom example. In Languages, the teacher identifies the number and type of errors in a familiar paragraph without correcting each location. Pupils use the model and criteria to find, explain and repair them. When a pupil lacks the relevant rule, the teacher gives direct teaching rather than leave the student in an unproductive search.

Top tips that make the technique work

Use detective work when pupils can solve it. Give the number and type of issues, then let students find and fix them.

Add the minimum cue. Narrow to a paragraph, step or criterion if the search is too broad.

Give direct teaching where needed. An error caused by missing knowledge is not a productive puzzle.

Protect action time. Plan redrafting, rehearsal, reattempting or re-testing into the lesson rather than hoping students act later.

Require an explanation. Ask what changed and why, then use a related task where the corrected learning matters.

Do and avoid

Do	Avoid
Set a bounded, achievable improvement action.	Cryptic feedback on an error students cannot diagnose.
Keep pupils doing the cognitive work.	Colour-coded corrections with no new thinking.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Feedback as Detective Work | Barton - Correcting in green pen does not work

Feedback follow-up

Checking improvement on a new task

Use feedback follow-up to find out whether the learning improved beyond the original correction. A new attempt gives more useful evidence than a rewritten answer alone.

How to do it well

1 Identify the intended improvement. State the knowledge, decision or feature the feedback should change.

2 Prepare a related task. Retain the conceptual demand while altering the example enough to require application.

3 Deliver feedback and action time. Model or prompt the correction and require the student to practise the repaired idea.

4 Collect an independent attempt. Reduce access to the original solution and inspect both answer and reasoning.

5 Return after a delay. Use a later retrieval or application task and decide whether further explanation, rehearsal or support is necessary.

What this sounds like

“You have corrected the worked example. Now close it and try this new case. Explain the decision that prevents the same error.”

Classroom example. After feedback on a misconception, pupils try a fresh question with comparable demand and no visible original solution. The teacher inspects the reasoning, not just the final answer. The learning returns in a later quiz. This separates immediate successful correction from the student’s ability to use the repaired idea independently over time.

Top tips that make the technique work

Plan the new item before feedback. This keeps the improvement focus precise and prevents the follow-up becoming an afterthought.

Keep the demand comparable. An unexpectedly harder question makes it difficult to interpret whether the correction helped.

Change enough to require thinking. A clone with one number swapped may reward imitation.

Revisit after a delay. Immediate success can depend on recent memory of the teacher’s explanation.

Inspect the process. A correct final answer may still hide the original faulty reasoning.

Do and avoid

Do	Avoid
Use independent application on a fresh task.	Counting a copied model as improved understanding.
Record remaining gaps for later teaching.	Never returning to the learning after the feedback lesson.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

Barton - Correcting in green pen does not work | Barton - Whole-Class Feedback

Worked examples

Showing a complete solution

Use a worked example when students need to see the full structure of a successful response before attempting a complex task. Make the important decisions and criteria visible.

How to do it well

1 Clarify the task. State what successful performance looks like and confirm the prerequisites students need.

2 Present the complete example. Show the response in a clear sequence or structure, with relevant workings and criteria.

3 Explain consequential choices. Point to where the method, evidence or feature satisfies the task and why an alternative would fail.

4 Rehearse the structure. Ask students to explain a step, compare a near-miss or complete a partly worked related example.

5 Hand over the method. Give a fresh independent task and remove the model when students can use the underlying structure without it.

What this sounds like

“This is a complete answer. Notice the claim, evidence and explanation. We will identify what each part does, then complete a different answer with the final section missing.”

Classroom example. A Business Studies worked example shows the complete response structure, calculations and interpretation. The teacher identifies why each part is needed, then contrasts a near-miss. Pupils complete a different example with the final section missing before attempting the whole response independently. The complete model provides a route into the demand.

Top tips that make the technique work

Choose a representative example. The model should expose the method students need, without an exceptional trick or distracting complexity.

Include the reasoning. A correct sequence with no rationale can encourage superficial copying.

Keep the layout transferable. Use a clear structure students can apply to another response.

Compare with a near-miss. A small error can reveal why the successful step is necessary.

Bridge to completion problems. Remove selected final steps before expecting the whole process independently.

Do and avoid

Do	Avoid
Show the complete method and its key decisions.	A polished answer with no explanation of its structure.
Move to a distinct related example.	Assuming copying the model establishes independent competence.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Worked Examples | Barton - I Do

DistillED Provide models

The visualiser

Discussing and improving student work

Use the visualiser to make student work a shared learning resource. Select a response because it illustrates a useful success, misconception or improvement decision.

How to do it well

1 Select the response. Choose a useful exemplar or near-miss and identify the learning feature it will reveal.

2 Prepare the display. Check readability, crop or mask irrelevant details and frame the sharing respectfully.

3 Ask students to inspect. Give silent time and a specific question about the criterion before taking contributions.

4 Model improvement. Annotate or edit the relevant section live, explaining the effect of each important change.

5 Require transfer. Students apply the same criterion to their own work or a fresh example, then you sample the resulting improvement.

What this sounds like

“This paragraph gives a clear claim. Let us find where the explanation stops short. We will improve that link, then you will inspect the same feature in your own paragraph.”

Classroom example. A teacher projects an anonymised paragraph that has a strong claim but a weak explanation. Pupils inspect the criterion, then watch a live edit and explain why it improves the link. Everyone makes an equivalent improvement in their own response. The projected work creates shared learning rather than a public judgement of its author.

Top tips that make the technique work

Select with a purpose. Decide which feature the class should notice before projecting the work.

Establish a respectful culture. Sharing should feel routine; anonymise where useful and avoid public embarrassment.

Keep the discussion narrow. One paragraph, diagram or process step often produces a clearer learning focus than the whole page.

Show the edit live. Explain why the change improves the response, retaining successful features.

Transfer to students’ own work. Everyone should make an equivalent improvement rather than merely watch the projected edit.

Do and avoid

Do	Avoid
Make the criterion and revision visible.	Displaying work solely to celebrate presentation.
Use contrasting responses to clarify quality.	Critiquing a named pupil rather than the response.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - The Visualiser The Complete Guide | DistillED - Whole Class Feedback

Clear instructions and front-loading

Specify participation before the task

Use front-loading when students begin responding or moving before they have heard the participation instructions. Say how to respond before giving the content that will capture attention.

How to do it well

1 Plan the action. Define the response location, participation mode, stopping point and what students should do if finished or stuck.

2 Secure attention. Use the shared signal and wait until students can access the instruction.

3 Front-load participation. State how students will respond before presenting the interesting question, object or task.

4 Check receipt. Ask a student to state the first action or a key restriction, clarifying briefly where needed.

5 Release clearly. Use the agreed cue, display any useful reminder and observe the first actions before moving into individual support.

What this sounds like

“In silence, write one answer on your board and keep it hidden. What are the two response rules? Now the question: which source is more reliable?”

Classroom example. Before displaying a diagnostic question, the teacher states that pupils will choose silently on boards and keep answers hidden. A listening check retrieves those rules. The question is then shown and the class commits privately. The interesting content has not arrived before the participation instructions, reducing unwanted calling out and contaminated responses.

Top tips that make the technique work

Put participation first. “On your board, silently…” comes before the question, not after an answer has been called out.

Keep the instructions numbered. Students can retrieve the first action and the checking point more easily than a long paragraph of directions.

Use an Instruction Sandwich. State the instruction, check that students heard it, then restate the concise version before release where useful.

Display complex directions afterwards. A stable reminder can support action without competing with the original spoken message.

Use a clear release cue. “When I say go” separates listening from movement and prevents the first instruction being lost.

Do and avoid

Do	Avoid
Specify where, how and when students respond.	Asking the question and adding instructions afterwards.
Check listening before students act.	Repeating every direction automatically without finding out what was missed.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Front-load the means of participation

Habits of attention

Teaching purposeful listening routines

Use habits of attention to teach where students should direct attention during explanation, modelling and discussion. Each habit needs a learning purpose and a recognisable cue.

How to do it well

1 Explain the learning purpose. Connect the habit to hearing a step, following a model, respecting a speaker or reducing distraction—not obedience for its own sake.

2 Select the highest-value behaviour. Teach only what matters in the current phase, such as tracking a model during explanation or orienting to a peer during discussion.

3 Model and rehearse briefly. Show what the habit looks and sounds like, practise once and return quickly to the learning.

4 Reinforce with short cues. Prompt early and quietly, narrate success occasionally and avoid turning the routine into constant commentary.

5 Apply selectively and adapt the form. Turn the habit off when pupils should read or write; make reasonable adjustments while preserving the attentional purpose.

6 Fade towards independence. Reduce verbal prompts to a gesture or question so pupils increasingly manage their own attention.

What this sounds like

“For this demonstration, put your pen down and attend to the diagram. Afterwards I will ask you to explain what changed between the two stages.”

Classroom example. In Drama, pupils shift from performance rehearsal to watching a demonstration. The teacher specifies the feature to attend to, then asks pupils to recall the key action and apply it in rehearsal. The attention habit supports access to the demonstration; visible posture alone is not treated as evidence that the performance technique was learned.

Top tips that make the technique work

Explain the purpose. A listening or viewing action is useful because it supports access to the idea, not because it looks compliant.

Name the relevant focus. Attention may belong on the model, speaker, diagram or text at different moments.

Practise the transition. Students need to switch attention fluently between writing, talk and shared explanation.

Check the message. Recall or application tells you more than posture, silence or nodding.

Avoid a rigid posture test. Use accessible ways of attending and do not treat direct eye contact as the only valid evidence.

Do and avoid

Do	Avoid
Teach the action and its purpose.	Enforcing a posture checklist for its own sake.
Make the change of attentional focus explicit.	Expecting attentional routines to compensate for unclear explanation.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Habits of Attention

Classroom transitions

Moving smoothly between learning activities

Use a taught transition to move students from one activity to the next without losing attention or learning time. Specify how to stop, shift resources and begin the new demand.

How to do it well

1 Teach it before using it. Name the transition, explain why it matters, model an example and non-example, and rehearse the sequence with feedback.

2 Precorrect immediately before the moment. Remind pupils what will stop, move, remain in place and begin; keep the cue brief and positive.

3 Reinforce the exact success. Describe the efficient, calm behaviour worth repeating rather than offering generic praise.

4 Actively supervise. Stay visible and mobile, scan likely breakdown points and avoid using transition time for administration.

5 Revisit and reteach. Refresh after holidays, timetable changes, new pupils or early signs of routine drift; distinguish an unlearned skill from inconsistent performance.

What this sounds like

“In ten seconds, stop the discussion. Keep the chairs where they are, face the board and wait for the next instruction.”

Classroom example. A Product Design transition moves pupils from drawing to collecting materials. The teacher primes the stop, gives one resource action at a time and rehearses the route beforehand. Students know where to place unfinished work and when to start again. Practical logistics do not compete with the explanation of the next learning task.

Top tips that make the technique work

Teach the transition as content. Model and rehearse the exact sequence before relying on it.

Prime the stopping point. Warn students what will happen and what their first next action will be.

Reduce simultaneous demands. Equipment movement, instructions and new subject content should not all compete at once.

Diagnose the failure. Students may not know the routine, or may know it but not yet perform it fluently.

Rehearse the troublesome component. A brief practice of resource return or partner rotation can solve more than a general reminder.

Do and avoid

Do	Avoid
Use clear stop and start cues.	Assuming the routine is obvious.
Protect the first moments of the new task.	Giving new academic instructions while equipment is moving.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Classroom Transitions

Precorrections

Preventing predictable problems before they start

Use a precorrection when a predictable situation repeatedly produces the same difficulty. Prompt the desired action immediately before that moment, using the pattern you have noticed.

How to do it well

1 Spot the pattern. Identify when, where, with whom and after what trigger the behaviour reliably occurs; act on a pattern, not an isolated bad day.

2 Name the replacement behaviour. Describe what pupils should do in positive, specific and observable language that is incompatible with the problem behaviour.

3 Adjust the setup. Change seating, resources, timing, task design or the physical environment so the expected response becomes easier.

4 Rehearse and prompt at the right time. Ask pupils to recall the routine shortly beforehand, then use a brief word, gesture or visual cue at the point of action.

5 Reinforce and check. Name successful enactment, monitor whether the pattern changes and reduce the prompt from a sentence to a gesture to nothing as fluency grows.

What this sounds like

“Before the trays move, carry yours with both hands and walk directly to your bench. We practised this route; use it now.”

Classroom example. A teacher notices that a familiar equipment transition repeatedly produces crowding. Just before it, the teacher reminds pupils of the taught collection route and adjusts the location of the resources. Successful execution is recognised. As the route becomes fluent, the spoken reminder is shortened and eventually removed.

Top tips that make the technique work

Be specific about the pattern. Identify when, where and what goes wrong so the reminder is targeted.

Name the action you want. “Carry the tray with both hands” gives more useful information than “Don’t mess about.”

Time the cue carefully. A reminder well before the risky moment may no longer be available when needed.

Adjust the setup too. Resource placement or task clarity can remove a recurring obstacle.

Fade the reminder. As the routine becomes fluent, reduce spoken prompts to a gesture or no prompt.

Do and avoid

Do	Avoid
Prompt a taught action just before it is needed.	A threat presented as a reminder.
Reinforce the successful response.	Repeating a generic warning without identifying a pattern.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Precorrections

Making participation safe

Prepare, rehearse and respond warmly

Use safe participation routines when fear of public error prevents students from contributing. Pair challenge with preparation and respond to uncertainty in predictable, respectful ways.

How to do it well

1 Build structural foundations. Establish predictable lesson routines, respectful listening agreements and a shared understanding that errors are information for learning.

2 Prepare before public response. Give thinking time, jotting, a model, guided practice or paired rehearsal so every pupil has something meaningful to contribute.

3 Make participation predictable. Front-load how pupils will respond, use warm cold call and structure the size or sequence of the public contribution.

4 Respond supportively to attempts and errors. Acknowledge the useful starting point, clarify the task or thinking and invite refinement without judging the pupil's ability.

5 Maintain challenge and accountability. Use prompts, second chances and gradual release to enable success; do not protect pupils by excluding them from the central thinking.

What this sounds like

"Everyone will have time to think and rehearse. I will ask different people to share. If an answer is uncertain, we will work with it and give you another chance."

Classroom example. A teacher prepares pupils with a written first thought and partner rehearsal before public questioning. When an answer is uncertain, a predictable prompt and later return provide another chance. Peer ridicule is addressed immediately. Students continue to face demanding questions, with a classroom routine that makes revealing uncertainty safer.

Top tips that make the technique work

Prepare before public response. Silent thinking, writing and paired rehearsal give students something to contribute.

Make error responses predictable. Explain how you will prompt, model or return after an uncertain answer.

Preserve challenge. Safety means students can attempt demanding learning without humiliation, rather than avoid participation.

Deal with ridicule promptly. Peer laughter or dismissive comments undermine the willingness to reveal thinking.

Recognise the learning move. Praise useful reasoning, revision or persistence precisely rather than celebrate confidence alone.

Do and avoid

Do	Avoid
Pair high expectations with preparation.	Cold calling as a public trap.
Treat errors as useful information.	Allowing uncertainty to become permanent opting out.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Making Participation Safe

Everyone–Together–Anyone

Planning participation for the whole class

Use Everyone–Together–Anyone to plan how all students think, rehearse and remain accountable. Choose a participation sequence that serves the learning rather than rely on confident volunteers.

How to do it well

1 Everyone thinks. Begin with silent solo thinking or Everybody Writes when the question requires an original response.

2 Together rehearses. Give partners a precise prompt, a short time limit and a listening expectation.

3 Anyone shares. Cold call individuals to report their own answer, their partner's reasoning or a refined joint response.

4 Everyone remains accountable. Invite pupils to compare, restate, improve or show a final independent response.

What this sounds like

"Everyone write a first explanation. Together, compare the evidence with your partner. Anyone may then be asked to explain which idea you kept and why."

Classroom example. A lesson first asks every pupil to write a proposed explanation. Partner talk then compares the evidence, and selected pupils share their reasoning. The teacher follows with an individual check where diagnosis matters. The participation sequence involves everyone without confusing a successful discussion with proof of each pupil's independent understanding.

Top tips that make the technique work

Begin with Everyone. An independent thought, jot or board response prevents the first speaker supplying everyone's answer.

Make Together purposeful. Partner talk should compare, explain or refine the thinking generated first.

Use Anyone for accountability. Select a prepared student to share and involve listeners in evaluating the response.

Choose the sequence for the task. Brief recall may use a different combination from extended reasoning; the framework is not a compulsory three-stage script.

Inspect thinking as well as activity. Busy talk or moving pens can still hide copied or shallow responses.

Do and avoid

Do	Avoid
Plan the cognitive action for every pupil.	Volunteer-only questioning.
Keep participation and diagnosis distinct.	Treating visible participation as proof of understanding.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Everyone, Together, Anyone

Second Chance / No Opt Out

Returning after ‘I don’t know’

Use Second Chance when a student’s first response is “I don’t know” or incomplete. Provide a route to a successful contribution and return to the student.

How to do it well

1 Set the expectation in advance. Explain that an uncertain pupil will receive more thinking time and be asked again after hearing one or two responses.

2 Pause and reframe. Acknowledge the hesitation without blame, state that you will return and give a precise listening task.

3 Invite useful peer responses. Ask one or two pupils for clear answers or reasoning that the original pupil can evaluate—not a long parade of guesses.

4 Return and require retrieval. Ask the original pupil to paraphrase, select the strongest response, explain why or add an idea of their own.

5 Reinforce persistence and improve. Acknowledge that the pupil stayed with the thinking, then give a task-focused prompt that increases precision next time.

What this sounds like

“I will ask two more people, then return to you. Listen for which explanation uses the key term accurately. You will tell us which one you chose and why.”

Classroom example. A pupil cannot initially explain the selected method. The teacher asks the pupil to listen to two alternatives, then returns for a choice and reason. A further related example follows where useful. The return keeps the pupil engaged in thinking while support makes the expected contribution attainable.

Top tips that make the technique work

Signal the return. The student should know they will have another chance and what to listen for.

Give a meaningful listening task. Ask them to select, paraphrase or develop a useful answer rather than repeat sounds mechanically.

Provide help when knowledge is missing. A prompt or model may be necessary before a successful response is possible.

Keep the tone warm. The purpose is continued participation and learning, not exposing hesitation.

Ask for application where appropriate. A further related question can distinguish hearing another answer from understanding the idea.

Do and avoid

Do	Avoid
Return after preparation or support.	Moving on and never returning.
Require an attainable, meaningful contribution.	Repeatedly demanding an answer without giving access to it.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Second Chance

Oracy for learning

Teaching speaking and listening explicitly

Use oracy routines to teach speaking and listening as skills that support disciplinary thinking. Clarify whether students are exploring an idea or presenting a developed explanation.

How to do it well

1 Build stable routines. Use predictable partners and explicit listening norms so transitions are quick and expectations are equitable.

2 Protect thinking before talking. Give pupils silent time to retrieve, annotate, select evidence or draft a first sentence.

3 Structure the cognitive question. Specify what pupils must explain, compare, justify or improve—not merely that they should discuss.

4 Provide the language or stimulus. Use a concrete example, text, image, sentence stem or target vocabulary when it helps pupils express the same ambitious idea.

5 Create listening accountability. Ask partners to restate, challenge or build, then cold call pupils to share their own or their partner's thinking.

What this sounds like

"This is exploratory talk. Offer an idea, ask a question and revise your explanation if the evidence changes your view. Afterwards, write the conclusion you can justify."

Classroom example. In Social Sciences, pupils use exploratory talk to test an explanation, ask for evidence and revise a tentative claim. A later presentational response requires a coherent conclusion using disciplinary vocabulary. The teacher models the listening and questioning moves, so talk quality is taught rather than assumed from the presence of discussion.

Top tips that make the technique work

Teach the language. Sentence stems and key vocabulary can help students articulate a complex relationship without supplying the content.

Make listening active. Partners can restate, build, question or challenge using evidence.

Distinguish exploratory from presentational talk. Tentative ideas and a polished explanation have different purposes.

Rehearse a specific talk move. Practise giving evidence or disagreeing respectfully before expecting an extended discussion.

Follow with writing where useful. A short revised explanation can consolidate what the talk clarified.

Do and avoid

Do	Avoid
Teach both speaking and listening explicitly.	Assuming fluent speakers already reason well.
Connect talk to the subject learning.	Unstructured conversation without a cognitive purpose.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Oracy

Everybody Writes / Silent Solo

Thinking in writing before discussion

Use Everybody Writes or Silent Solo when students need to form their own response before discussion. Writing gives quieter students a prepared idea and makes initial thinking visible.

How to do it well

1 Set the cognitive prompt. Choose a question requiring a claim, explanation, comparison or decision that students have the knowledge to attempt.

2 Specify the product. State how much to write and the important feature, so preparation does not become an unbounded essay.

3 Protect independent time. Students write silently without seeing another response or a model answer.

4 Use the written thinking. Compare in pairs, sample responses or project a useful contrast while keeping listeners accountable.

5 Revise deliberately. Students improve their own response and identify the reason for a change, preserving the learning gained from discussion.

What this sounds like

“Write your own claim and one piece of evidence before speaking. Compare with your partner, then revise the explanation in a different colour and explain the change.”

Classroom example. Before debating a historical claim, every pupil writes a position and supporting evidence. Partners compare the reasoning, then pupils revise their own statement and explain the change. The written first attempt protects independent thought; the final revision captures what the discussion added instead of leaving the learning inside an unrecorded conversation.

Top tips that make the technique work

Keep the first write bounded. A claim, two reasons or a first decision is often enough to prepare discussion.

Protect silence. The purpose is individual generation before a partner’s response influences the starting idea.

Choose a prompt worth writing. A trivial question may not need this preparation; reasoning benefits more.

Use the writing afterwards. Ask students to compare, refine or explain their first response rather than abandon it.

Build read–write–discuss–revise. The final revision captures what the discussion added and makes changed thinking visible.

Do and avoid

Do	Avoid
Require everyone to commit an initial idea.	Allowing partner discussion during the individual stage.
Connect writing to later talk and revision.	Asking for writing that never informs the next task.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

Barton — Turn and Talk | Barton Independent practice

Explaining with precision

Plan, sequence and rehearse the explanation

Use precise explanation when the content is correct but the message is too long, unclear or difficult to follow. Plan what to say, what to show and what to leave out.

How to do it well

1 Define the core idea. Write the smallest accurate explanation that makes the intended relationship clear.

2 Sequence the components. Introduce necessary terms and prerequisites before adding the dependent idea; use a concrete example.

3 Align the representation. Plan the diagram, worked step or demonstration and where students should look at each moment.

4 Deliver and pause. Use concise language, check listening and allow reading or processing before introducing another component.

5 Capture and rehearse. Leave a useful summary, guide a related attempt and repair the precise point the responses reveal.

What this sounds like

“This arrow shows the transfer, not the store. Watch the arrow while I explain the movement. Now tell me what the arrow represents before we consider a new example.”

Classroom example. A Science teacher rehearses a concise explanation of a particle relationship, chooses a simple diagram and plans two pauses. Students first retrieve a key statement, then apply it in a new situation. A common error leads to a focused contrast. The explanation remains precise rather than expanding into several distracting related topics.

Top tips that make the technique work

Rehearse the difficult sentence. A brief rehearsal often exposes vague terms, unnecessary detours and an overloaded sequence.

Funnel attention. Align your words, pointing and visual display so students know which feature matters now.

Use consistent terminology. Switching casually between labels can create extra interpretation work for novices.

Capture the key idea. A stable sentence, diagram or checklist supports early practice after the spoken explanation ends.

Check receipt and application separately. A listening check and a fresh example answer different questions about the explanation.

Do and avoid

Do	Avoid
Plan a concise sequence and useful representation.	Adding interesting detail that distracts from the main idea.
Pause between coherent chunks.	Reading dense slide text aloud while students try to read it.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Explaining With Precision

Cognitive simplicity

Designing slides and resources that support attention

Use cognitive simplicity when slides or resources compete with the explanation. Reduce unnecessary searching, reading and distraction so students can attend to the content that matters.

How to do it well

1 Identify the essential information. Decide what students need to see, hear or read at the current point.

2 Remove competing material. Delete decorative elements, irrelevant detail and duplicated language that adds no explanatory value.

3 Reduce searching. Group related content, label clearly and keep the important feature easy to locate.

4 Sequence the reveal. Introduce information in coherent chunks and separate reading, listening and copying when they would compete.

5 Test the resource. Check readability from the back of the room and whether students can locate the reference needed during practice.

What this sounds like

“Read these two sentences silently first. I will then explain the distinction. Watch the highlighted phrase as we connect it to the diagram.”

Classroom example. A teacher removes an unrelated image, moves labels beside the diagram and reveals one part at a time. Students read a short definition in silence before the teacher explains its connection to the visual. The revised resource reduces competing demands while retaining the reference information pupils need during practice.

Top tips that make the technique work

Remove decorative competition. An irrelevant picture or animated effect can attract attention away from the learning feature.

Place related information together. Keep a label beside its diagram feature and a condition close to the step it affects.

Reveal deliberately. A completed diagram or crowded page may demand more processing than the current explanation requires.

Stop talking over new reading. Give students time to inspect substantial unfamiliar text before explaining it.

Preserve useful reference material. Simplicity does not mean removing the model or vocabulary students need to practise successfully.

Do and avoid

Do	Avoid
Use layout and sequence to direct attention.	Decorative images, clutter and redundant text.
Keep text and visuals purposeful.	Removing essential information merely to make a slide look sparse.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

DistillED — Explaining With Precision

Live drawing

Building a representation step by step

Use live drawing to build a representation in the same sequence as the explanation. Clarity and purpose matter more than artistic polish.

How to do it well

1 Connect and display. Start with a blank surface, confirm visibility and link the representation to relevant prior knowledge.

2 Draw and label gradually. Narrate the concept as each element appears; keep the visual simple and essential.

3 Pause and read. Stop speaking while students read labels or annotations so verbal and visual information do not compete.

4 Explain and point. Trace arrows, point to the current feature and check understanding before adding more.

5 Recreate and rehearse. Students copy, label, predict or explain the representation so the task is active rather than observational.

What this sounds like

“I will add one arrow at a time. Explain what this connection represents before I draw the next one. Then complete the missing part on your own diagram.”

Classroom example. In Geography, a teacher draws a simple process diagram in the order events occur, adding arrows and labels gradually. Pupils explain each connection before completing a missing stage independently. The drawing makes the causal sequence visible; students are not judged on reproducing an artistic copy of the finished diagram.

Top tips that make the technique work

Start with the meaning. Decide which relationship the diagram will explain before deciding how attractive it should look.

Rehearse the strokes. Plan the order so the emerging drawing supports the spoken sequence.

Label gradually. Keep words close to the relevant part and pause while students read them.

Use pointing and tracing. Show how an arrow, boundary or connection represents the concept.

Ask students to recreate meaning. They can complete a missing part or explain a relationship rather than merely copy the finished picture.

Do and avoid

Do	Avoid
Build the diagram in coherent stages.	Complex artwork that hides the concept.
Rehearse the relationship it represents.	Drawing every label before students know what to attend to.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Live Drawing

Storytelling and analogies

Making an explanation coherent and memorable

Use a story or analogy when it clarifies a sequence, relationship or abstract idea. Keep the narrative tied to the disciplinary meaning students must learn.

How to do it well

1 Identify the relationship. Decide what the story or comparison will make easier to understand.

2 Select a familiar structure. Use a short sequence or analogy students can follow without extensive new background.

3 Explain the mapping. Link the important narrative features to the subject concept and show the cause or relationship.

4 State the boundary. Identify a way the analogy differs from the target idea before that difference becomes a misconception.

5 Translate back. Ask students to explain the concept in disciplinary terms and apply it to an example outside the original story.

What this sounds like

“Think of this as a pressure cooker building tension. In the text, what represents the pressure? Where does the analogy help, and where does it stop being accurate?”

Classroom example. An English teacher uses a short analogy to explain mounting tension in a text. Each part of the comparison is mapped to evidence, and a limitation is named. Pupils then explain the tension using subject vocabulary and quotations. The analogy helps access the idea without becoming a substitute for precise textual analysis.

Top tips that make the technique work

Keep cause and consequence visible. A story should help explain why an event or change occurs, not merely entertain.

Map the analogy explicitly. State what each part represents in the target idea.

Name the limits. Tell students where the familiar comparison stops being accurate.

Remove unnecessary detail. Interesting characters or episodes can become the remembered feature instead of the concept.

Return to subject language. Students need to explain the idea without relying on the story alone.

Do and avoid

Do	Avoid
Choose a narrative that carries the intended explanation.	A memorable story with an unclear learning point.
Connect it back to precise vocabulary.	Treating an analogy as literally identical to the concept.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

DistillED — Storytelling

Micro-decisions

Notice, interpret and respond in the moment

Use micro-decisions to make teaching responsive while the lesson is unfolding. Notice a useful signal, interpret its likely cause and choose a proportionate next action.

How to do it well

1 Notice. Attend to responses, work, hesitation, explanation quality and participation. Deliberately look for pupils whose thinking is not yet visible.

2 Interpret. Decide whether the evidence shows a task problem, a process gap, insecure prior knowledge, a misconception or a self-regulation barrier.

3 Respond. Probe, continue, slow down, re-model, reteach, change the participation structure, alter support or increase challenge according to the diagnosis.

4 Check. Sample understanding again. A micro-decision is incomplete until there is evidence that the action closed the intended gap.

What this sounds like

“I can see correct answers, but the explanations use different rules. I will ask for the reasoning before deciding whether we are ready to continue.”

Classroom example. A correct board response is accompanied by an incorrect explanation. The teacher probes the decision, notices the same pattern elsewhere and models the crucial distinction. A fresh check tests whether the reasoning changed. The small instructional adjustment follows evidence rather than the planned pace or the reassuring appearance of correct answers.

Top tips that make the technique work

Know what you are looking for. A planned misconception or key decision makes observation more useful than a general scan for engagement.

Probe before changing course. A correct answer may need reasoning; a wrong answer may be a slip rather than a shared misconception.

Find the missing thinking. Ask whose response is absent before judging the whole class from the visible few.

Make the smallest effective change. A cue, shorter step or clearer representation may be sufficient without restarting the lesson.

Check whether it landed. The next response should test the issue that prompted the adjustment.

Do and avoid

Do	Avoid
Link the action to an observed pattern.	Changing course because of one unrepresentative response.
Recheck the affected learning.	Persisting with the plan despite a clear common gap.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Micro-Decisions

High success rates

Deciding when students are ready for independence

Use success-rate evidence to judge whether practice is productive and students are ready for greater independence. Interpret accuracy alongside reasoning, task demand and the distribution of errors.

How to do it well

1 Sequence for early success. Teach one manageable component at a time and anticipate misconceptions before complexity is added.

2 Model and guide. Make an accurate first example visible, then co-construct and rehearse several short examples with high participation.

3 Check accuracy and reasoning. Use all-pupil responses, short quizzes or sampled work to estimate the class success rate and the pattern of errors.

4 Adjust support or challenge. Below the productive range, re-model, shrink the step or restore a scaffold. When accuracy is consistently above about 90%, fade support or add depth, variation or transfer.

5 Reinforce and revisit. Use immediate correction, later retrieval and spaced practice so accurate responses become fluent and durable.

What this sounds like

“The answers are mostly accurate. Before you work independently, explain the first decision on a new example. I will check the remaining gaps while you practise.”

Classroom example. Most guided answers are correct, but several students have copied a visible method without explaining the first decision. The teacher uses a distinct example and collects reasoning before release. Secure pupils receive a useful variation while the remaining group gets targeted guidance. Readiness is judged from the demand and pattern, not the percentage alone.

Top tips that make the technique work

Treat percentages as signals. Around 80% can be a useful practical reference in some tasks, not a universal rule or guarantee of readiness.

Inspect what the errors concern. A small minority missing a crucial prerequisite still needs follow-up.

Distinguish accuracy from copying. Success with an identical example or visible answer can exaggerate competence.

Increase challenge when secure. Fade support, change a condition or require explanation instead of assigning more of the same.

Return to the learning. Initial success is a starting point for fluency and later retrieval, not proof of permanent mastery.

Do and avoid

Do	Avoid
Use accuracy, reasoning and task structure together.	Mechanical release at a fixed percentage.
Adjust support before repeated failure.	Making every practice task trivial to achieve high scores.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED — Obtain a High Success Rate

Atomisation

Breaking learning into prerequisite and new knowledge

Use atomisation in lesson preparation to identify the small pieces of knowledge and action a successful performance requires. Separate existing prerequisites from genuinely new learning.

How to do it well

1 Choose the endpoint. Write a representative question, performance or response that embodies the intended learning.

2 Unpack the actions. Work through it and list the knowledge, interpretations and decisions required at each point.

3 Separate old and new. Mark components that students should already know and those the lesson must explicitly teach.

4 Plan the prerequisite checks. Use short direct prompts, repair insecure foundations and recheck before introducing dependent content.

5 Build the episode. Sequence the new components, a concise model, guided rehearsal and recombination into an independent whole-task attempt.

What this sounds like

“To calculate this mass, students must interpret the formula, find the amount, use the relationship and manage units. Which are prerequisites, and which are new today?”

Classroom example. Preparing a gas-volume calculation, the teacher separates interpreting the question, converting units, recalling the relationship and using the amount correctly. A short check reveals that unit conversion is insecure. That prerequisite is repaired before the new application is modelled. Atomisation changes the teaching route rather than merely producing a longer planning list.

Top tips that make the technique work

Start with a real task. A representative example exposes the decisions students actually need to make.

Ask what each step depends on. Reading a symbol, selecting a rule and completing an operation may require different knowledge.

Check prerequisites separately. A complex final question can hide which small component is missing.

Avoid endless fragmentation. Keep the overall concept visible and recombine the pieces after rehearsal.

Use the analysis to plan teaching. Atomisation is useful when it changes the explanation, check or practice sequence.

Do and avoid

Do	Avoid
Identify prerequisite and new components explicitly.	Reducing the task to disconnected facts.
Plan a direct check for the likely bottleneck.	Assuming a difficult final answer reveals which component failed.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Atomisation

Walking-talking mocks

Attempt, diagnose, model and re-check

Use an active walking-talking mock to combine expert modelling with student attempts and diagnosis. Give explanation time where the responses show it is needed.

How to do it well

1 Attempt and commit. Work through one question at a time; students write an answer and confidence rating before any model is shown.

2 Gather evidence. Use boards for a final answer, first decision or line of working; allocate staff scanning zones where necessary.

3 Model responsively. Confirm secure work briefly or explain the specific difficulty, making expert decisions visible.

4 Recheck and stretch. Use a fresh equivalent question after teaching, then a What If variation where the underlying idea is secure.

5 Create Review Cards. Begin with high-confidence errors, record the mistake and corrected example, and schedule later self-quizzing.

What this sounds like

“Try this question first and rate your confidence. Show the first line of working. We will model the part causing difficulty, then everyone will attempt a related question.”

Classroom example. During mock review, pupils attempt a question and rate confidence before staff scan boards in zones. A common first-step error receives a short model and a fresh shadow question. Secure answers receive a purposeful What If variation. Pupils then build Review Cards from selected errors instead of leaving only with an annotated paper.

Top tips that make the technique work

Attempt before the model. Students need a chance to make decisions before seeing the teacher’s solution.

Commit confidence before feedback. A score alongside the answer helps identify confidently held errors for later review.

Scan in zones. Where several staff are present, assign areas so every response can be seen.

Use equivalent follow-ups. A shadow question can check the repaired idea without students simply copying the original.

Make the revision usable. Review Cards give students a question they can test themselves on later.

Do and avoid

Do	Avoid
Use responses to decide explanation time.	Teacher talk through the whole paper before pupils attempt it.
Include a recheck and later retrieval.	Giving every question identical treatment regardless of need.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Walking-Talking Mock Exams

Review Cards

Turning errors into future retrieval

Use Review Cards to turn corrected errors into a personal retrieval resource. Separate the question from the solution so students can attempt it again without seeing the answer.

How to do it well

1 Select the error. Choose a consequential gap from a quiz, assessment or mock, prioritising a confidently held mistake where useful.

2 Write the front. Copy a complete question or prompt that can be answered independently.

3 Build the back. Record the topic, original mistake, corrected method and annotations that explain the crucial decision.

4 Add support. Include a specific reference for help or further practice, checking that the solution is accurate.

5 Use the deck. Attempt the front before checking the back; revisit incorrect cards sooner and return to secure cards after a longer interval.

What this sounds like

“Put the question on the front. On the back, record the topic, your mistake, a corrected worked example and the note that would help you avoid it next time.”

Classroom example. A pupil turns a confidently incorrect quiz item into a card. The front contains the question; the back explains the mistake, corrected method and where to find more practice. Later, the pupil attempts the front before turning it over. An incorrect attempt returns sooner in the deck rather than disappearing after the original correction.

Top tips that make the technique work

Begin with a high-confidence error. It identifies an important mismatch between certainty and correctness.

Annotate for the future self. Include why the decisive step works and what caused the original mistake.

Add a help reference. Link to an appropriate worked example, revision resource or practice location.

Quiz before turning over. Looking at the solution first turns the card into rereading.

Return errors sooner. A simple deck routine can revisit incorrect cards before secure ones without needing a complex system.

Do and avoid

Do	Avoid
Keep the question and solution on opposite sides.	A card containing only a final answer.
Revisit the cards through independent attempts.	Making cards that students never use for retrieval.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Walking-Talking Mock Exams: Review Cards | Barton Low-stakes quiz

‘What if?’ questions

Vary a condition to deepen understanding

Use What If questions to deepen a secure example by changing one condition. Students should adapt the reasoning they have established rather than begin an unrelated problem.

How to do it well

1 Establish a secure example. Check the class can complete and explain the basic method or idea.

2 Inspect the structure. Identify the steps or criteria that matter, using the model where useful.

3 Alter one condition. Change a value, representation, requirement or constraint that makes students adapt their reasoning.

4 Predict and test. Ask everyone to explain the expected effect before checking a calculation or completing the altered task.

5 Reset or extend. Use a clean related example for independence, or a further variation for depth; do not force every stage when it adds little.

What this sounds like

“What if this value doubles? Before calculating again, explain which part of your working changes and which part stays the same.”

Classroom example. After a secure worked example, the teacher alters one condition and asks pupils to predict the consequence. Pupils explain which step changes and which relationship stays valid. A clean related task then checks independent setup. The variation deepens the established idea without introducing several new demands at once.

Top tips that make the technique work

Change as little as necessary. A small consequential alteration makes the relationship easier to inspect.

Ask for a prediction first. Students can state what they expect to change and what will remain true.

Use Tick Trick thoughtfully. Compare working with the model to identify the decisions or criteria that earn credit.

Reset when needed. A clean, distinct example tests whether students can start the method independently.

Extend after security. A demanding variation is more useful when the basic method is sufficiently understood.

Do and avoid

Do	Avoid
Preserve the underlying concept.	Changing several unfamiliar demands simultaneously.
Ask for the reason the answer changes.	Using extension while the original decision remains insecure.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — We Do | Barton Walking-talking mock exams

Golden Silence and active observation

Launching independent practice

Use Golden Silence and active observation to protect the launch of independent practice. Establish the working norm, then alternate purposeful circulation with a view of the whole class.

How to do it well

1 Prepare the launch. Give clear instructions, a stuck protocol and a useful finish task, then check listening before release.

2 Protect Golden Silence. Use a brief period of visible whole-class observation to establish independent effort.

3 Circulate purposefully. Inspect a specific feature of early work along a planned route and give short targeted feedback.

4 Step back and decide. Regain a view of the room, identify participation gaps and distinguish individual need from a common error.

5 Continue or reset. Preserve productive practice, use a brief class repair when justified, or a planned discussion break followed by renewed independent work.

What this sounds like

“Begin independently. I will watch the first thirty seconds, then check first attempts. If you need help, identify the step that is unclear while continuing what you can.”

Classroom example. After checking instructions, the teacher protects a brief silent launch from a position with a whole-room view. Circulation then checks first attempts along a route, followed by a step back to inspect participation. A targeted prompt is revisited later. The teacher avoids spending the opening minutes absorbed at one desk.

Top tips that make the technique work

Protect roughly the first thirty seconds. Stand where you can see the room, recognise the expected start and correct unobtrusively.

Use 30:30:30 flexibly. A brief launch, circulation period and step-back observation can stop individual support absorbing all attention.

Eliminate blackspots. A planned route should reach every part of the room, not just raised hands.

Give one precise intervention. Prompt the next step, check it was heard and return after the student works.

Use 4–2 cycles selectively. A short independent burst followed by paired checking can sustain thinking when it serves the task; it is not a compulsory timer routine.

Do and avoid

Do	Avoid
Secure the start before extended individual help.	Circulating into a long conversation immediately.
Step back regularly to observe the whole room.	Treating silent appearance as evidence that everyone is thinking.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Maximising participation during independent practice

Confidence scores

Finding confidently held errors

Use confidence scores alongside answers to identify uncertainty and confidently held errors. Confidence adds context to performance; it does not replace evidence of knowledge.

How to do it well

1 Choose the use. Apply ratings to selected quiz or assessment answers where the information can guide feedback.

2 Explain the scale. Use clear endpoints and establish that honest uncertainty is helpful.

3 Attempt and rate. Students answer independently and record confidence before solutions or discussion.

4 Compare with accuracy. Identify confident errors, uncertain correct answers and insecure knowledge, then probe where needed.

5 Plan the response. Give relevant feedback and future practice, using high-confidence errors for Review Cards where appropriate.

What this sounds like

“Before we reveal the solution, rate how confident you are that your answer is correct. We will compare confidence with the evidence and choose what to revisit.”

Classroom example. A quiz answer is wrong despite a high confidence rating. The teacher uses that mismatch to prioritise the explanation and a Review Card. A correct but uncertain answer receives a reasoning check and further practice. Ratings provide useful context to observed performance, rather than a separate attainment measure.

Top tips that make the technique work

Commit before feedback. Ratings made after seeing the answer cannot reveal the original certainty.

Use a simple scale consistently. For example, explain what the ends of a 0–10 scale mean without turning every number into a complex rubric.

Prioritise confident errors. They are useful candidates for explanation, comparison and Review Cards.

Investigate low-confidence correct answers. The student may need fluency practice or may have guessed.

Keep the tone neutral. Honest uncertainty is more useful than students inflating ratings to appear successful.

Do and avoid

Do	Avoid
Record a rating beside an independent answer.	Treating confidence as attainment.
Use the mismatch to guide review.	Rating only after the correct answer is public.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — The role of confidence in learning

Learner-generated examples

Testing concept boundaries

Use learner-generated examples to test the boundaries of a concept. Constraints help students move beyond the first obvious example and expose what they think counts.

How to do it well

1 Choose the concept. Identify the defining feature students should apply and confirm they have encountered it.

2 Generate a first example. Give silent time and collect a bounded response before sharing suggestions.

3 Broaden the range. Ask for another valid example, then an unusual example or one satisfying an added constraint.

4 Test the boundary. Ask for a convincing non-example and an explanation of the feature that excludes it.

5 Compare and refine. Discuss selected cases, correct the definition where necessary and test a fresh constraint individually.

What this sounds like

“Give one example, another example and one that no one else is likely to choose. Finally, give a non-example that could fool someone, and explain the difference.”

Classroom example. In Languages, students generate a valid sentence, a second valid example and one satisfying an added constraint. They then write a plausible incorrect example and explain the rule it breaks. The range reveals whether pupils understand the grammatical boundary or rely only on the first familiar sentence pattern.

Top tips that make the technique work

Ask for several examples. A second or unusual example reveals more than a single familiar case.

Add purposeful constraints. Specify a feature that forces students to reconsider the concept rather than repeat their first answer.

Include a plausible non-example. Ask for a case someone might think valid and explain why it is not.

Inspect boundary reasoning. Students should justify membership using the definition rather than how unusual the example looks.

Teach the concept first. Generating examples is most useful when students have enough knowledge to test the constraints.

Do and avoid

Do	Avoid
Use constraints that expose the defining feature.	Unbounded example generation with no criterion.
Require explanations as well as examples.	Treating unusual answers as automatically sophisticated.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Learner-Generated Examples

Consolidation

Securing new learning before independent practice

Use consolidation after initial explanation and guided practice to secure the new idea before increasing independence or complexity. Make students reconstruct and explain the learning.

How to do it well

1 Select the central learning. Identify the principle or process students should retain from the episode.

2 Ask for reconstruction. Use a brief explanation, completion task or related attempt that requires students to recall and organise the idea.

3 Compare thoughtfully. Let students inspect a model or another response and explain meaningful differences.

4 Repair the insecure component. Re-model a specific gap and give another fresh attempt where necessary.

5 Connect the next stage. Move into purposeful independent practice or application when sufficiently secure, and plan a later retrieval return.

What this sounds like

“Without looking at the model, explain the first decision and complete a related example. Then compare your method and identify the step you need to practise.”

Classroom example. After guided practice, pupils reconstruct the key method and complete a short related task with less visible support. Comparison with the model reveals one insecure decision, which gets a focused repair. A later retrieval return is planned. Consolidation secures the episode without being treated as proof that the learning will remain accessible permanently.

Top tips that make the technique work

Keep it focused on the new learning. A broad topic recap may miss the specific decision taught in this episode.

Require an active response. Explanation, completion or a short independent attempt shows more than rereading the model.

Use the model deliberately. Let students compare initially, then remove it for a fresh attempt.

Do not equate consolidation with easy work. Students should think about the key relationship, not merely copy a summary.

Connect to later retrieval. Consolidation helps initial security; spaced returns are still needed.

Do and avoid

Do	Avoid
Reconstruct the important idea or process.	A final slide that students passively reread.
Use the response to decide readiness.	Assuming immediate success removes the need for later retrieval.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Consolidation

Problem-solving

Applying secure knowledge in less familiar tasks

Use problem-solving to help students apply secure knowledge when the route is less familiar. Plan the extra decision demand rather than disguise new teaching inside a complex task.

How to do it well

1 Identify the transferable knowledge. Confirm students can use the relevant concepts and procedures in familiar tasks.

2 Select the new demand. Change the context, representation or route choice while keeping the underlying content accessible.

3 Plan and represent. Students identify known information, the goal and a useful diagram or structure.

4 Attempt with purposeful support. Use prompts, comparison or modelling when the search reveals a specific barrier.

5 Compare and generalise. Discuss approaches and the clues that selected them, then apply the reasoning to another task.

What this sounds like

“What do we know, what are we trying to find and which relationship could connect them? Sketch the situation before selecting your approach.”

Classroom example. In Maths, pupils use familiar methods in a task where the approach is not labelled. They represent the problem, identify useful information and justify the selected route. When a barrier concerns a taught relationship, the teacher prompts or models that point. A final comparison explains why alternative routes do or do not work.

Top tips that make the technique work

Check the knowledge base first. Students cannot reason effectively with relationships or methods they have not learned.

Separate selection from execution. Ask which approach fits and why before completing every step.

Teach a useful heuristic explicitly. A diagram, simpler case or organised list can support the task without becoming a universal trick.

Model when search becomes unproductive. Sustained confusion may need guidance rather than more unguided time.

Compare routes after solving. Explain why different approaches work and when each is useful.

Do and avoid

Do	Avoid
Match the unfamiliar demand to secure knowledge.	Calling an untaught procedure problem-solving.
Discuss the reasoning behind the chosen route.	Valuing struggle without attending to whether it produces learning.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Problem-Solving

Purpose

Helping students understand what they are learning and why

Use a clear purpose to help students understand what the lesson is building towards and why the learning matters. Connect activities to the knowledge, performance or later use.

How to do it well

1 Define the destination. State the knowledge or performance students will develop, using language they can understand.

2 Select a concrete frame. Use an endpoint question, model or meaningful application to show what the learning enables.

3 Connect the first step. Explain how today's component contributes to the larger idea.

4 Keep the purpose visible in practice. Refer to it when choosing examples, checking a decision or explaining an improvement.

5 Return to the destination. Ask students to use the learning in the original question or a related task so the purpose has a concrete outcome.

What this sounds like

"We are learning how evidence supports a historical judgement. This paragraph gives us a way to compare explanations rather than list events."

Classroom example. A Product Design lesson frames material properties through a concrete selection problem. The teacher explains how each taught property helps justify the choice. Practice returns to that same decision with a different product. The purpose connects activity to disciplinary reasoning rather than an artificial claim that every detail has an immediate everyday use.

Top tips that make the technique work

Name the learning rather than the activity. "Explain the cause" is clearer than "Complete the worksheet."

Keep purpose concrete. A question, performance or problem can make the destination visible.

Show a genuine connection. Use the curriculum or subject application rather than an invented claim that every idea is needed in daily life.

Revisit at the decision point. Explain how the current step contributes to the endpoint.

Preserve curiosity without delaying instruction. A hook is useful when it directs attention towards the learning, not when it becomes a separate entertainment.

Do and avoid

Do	Avoid
Connect the task to a clear learning destination.	An activity label presented as the purpose.
Use a meaningful subject rationale.	A lengthy motivational introduction unrelated to the content.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Purpose

Retrieval homework

Revisiting learning beyond the lesson

Use retrieval homework to revisit taught knowledge between lessons. Make the independent attempt possible, provide useful feedback and connect gaps to future teaching.

How to do it well

1 Select the retrieval content. Choose important taught knowledge and procedures, including planned returns to older topics.

2 Keep the task manageable. State the response, approximate scope and how students should act when stuck.

3 Protect the attempt. Give instructions to work independently before checking solutions or a support reference.

4 Build corrective feedback. Provide accurate explanations or worked solutions and a way to record consequential errors.

5 Connect to the next lesson. Recheck selected items, address common gaps and schedule further retrieval where needed.

What this sounds like

“Try these questions from memory first. Then use the worked solutions to identify errors. Bring one question you need to revisit; we will check the main decisions next lesson.”

Classroom example. A Science homework mixes taught calculations and older retrieval. Pupils attempt before checking worked solutions and record one error to revisit. The next lesson uses a pivotal related question to sample the learning. The homework supports retrieval and feedback rather than becoming independent discovery of a procedure students have never been taught.

Top tips that make the technique work

Use taught, accessible content. Homework should not rely on students discovering a new method without instruction.

Separate attempt and solution. Students need a chance to retrieve before checking worked answers.

Mix recent and older learning. A deliberate sequence strengthens return to important knowledge across time.

Provide a help route. A specific model, resource or reference supports students when recall fails.

Use the result in class. Sample a pivotal item or misconception so homework informs later retrieval and explanation.

Do and avoid

Do	Avoid
Set homework through the current BSCS arrangements.	Untaught independent content as retrieval.
Use feedback and later retrieval.	Checking completion without attending to the learning.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Homework

Getting students on board with retrieval

Explain, demonstrate and build habits

Use an explanation and demonstration of retrieval when students see quizzes as judgement or prefer familiar rereading. Help them understand why attempting recall is part of learning.

How to do it well

1 Identify the concern. Find out whether students fear scores, do not understand the purpose or lack a useful retrieval routine.

2 Explain simply. Describe how trying to recall differs from rereading and why feedback follows the attempt.

3 Demonstrate with accessible content. Let students experience independent recall and checking without making the task a test of self-worth.

4 Establish the routine. Use predictable instructions, achievable demands and respectful responses to errors.

5 Reinforce the learning purpose. Show how previous gaps return in future practice and how students can use retrieval independently.

What this sounds like

“You may recognise this when reading it, but can you produce it without the page? Try first; the gap tells us what to practise, and the feedback helps repair it.”

Classroom example. A teacher explains why recognising a paragraph is different from producing its key idea without looking. Pupils try an accessible recall task, receive feedback and meet the same knowledge again later. The routine demonstrates a learning purpose and treats errors respectfully, helping build trust in retrieval beyond a single explanation.

Top tips that make the technique work

Explain the learning purpose. A low-stakes attempt strengthens access to knowledge and shows what to revisit.

Demonstrate a fair comparison. Use a simple experience to contrast recognition with generating an answer, without promising an instant universal result.

Normalise effort. Difficulty recalling is information about the learning, not a reason to avoid the attempt.

Make success attainable. Use well-taught content and a useful prompt where total failure would prevent productive practice.

Keep the routine consistent. Repeated experiences of attempt, feedback and improvement build more trust than a single motivational speech.

Do and avoid

Do	Avoid
Explain the purpose and give corrective feedback.	A public ranking that undermines the low-stakes message.
Use mistakes to guide future practice.	Claiming effortful failure is always beneficial.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Getting students on board with retrieval

Precise answers / Don’t round up

Helping students refine a response

Use precise-answer routines when a partly correct response is being accepted as complete. Help the student improve the answer rather than silently replacing it with your own explanation.

How to do it well

1 Listen to the actual answer. Judge what the student said against the intended knowledge rather than what you think they meant.

2 Name the useful start. Recognise the accurate element and identify the missing or imprecise feature.

3 Prompt a revision. Use a clue, key term, criterion or short rehearsal to make improvement achievable.

4 Return for the full answer. Ask the pupil to articulate the complete explanation, then involve listeners in restating or applying it.

5 Check a fresh case. Use a related question when needed to establish that the refined reasoning can be used beyond the rehearsed wording.

What this sounds like

“You have the main idea. Say it again using ‘concentration’ and explain which way the particles move. I will give you a moment to prepare.”

Classroom example. A pupil’s answer contains a correct idea but omits the condition that makes it valid. The teacher recognises the accurate start, prompts the missing feature and returns to the pupil for the full explanation. A new example follows. The teacher does not silently supply the condition and treat the original response as complete.

Top tips that make the technique work

Identify the missing feature. Decide whether the issue is vocabulary, evidence, a condition or the relationship between ideas.

Use an attainable prompt. Point to the needed feature without providing the whole answer when the pupil has the knowledge.

Return to the same student. They should articulate the completed response after the support.

Use “Say it again, better”. Ask for a fuller, more precise version and specify what must improve.

Keep the tone warm. Accuracy and a supportive response can coexist; do not turn refinement into public humiliation.

Do and avoid

Do	Avoid
Require the pupil to complete or refine the response.	Rounding up a partial answer as fully correct.
Make the criterion for precision explicit.	Supplying the explanation and crediting the pupil with it.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Don’t round up